
On Minimum Depth and Width of Floating-Point Neural Networks for Representing Floating-Point Functions

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Abstract

Research on the expressive power of neural networks has identified the minimum depth and width of neural networks that enable universal approximation and memorization. However, existing results are derived under exact arithmetic and cannot be directly applied to real implementations on computers, which can only use a finite set of numbers and inexact machine operations with round-off errors. In this work, we study floating-point ReLU networks that have floating-point parameters and use floating-point operations. Specifically, we investigate their minimum depth and width to represent all functions from the set of floating-point vectors $\mathbb{F}^d$ to the set of floating-point numbers $\mathbb{F}$. We first show that the minimum depth for representing all functions from $\mathbb{F}^d$ to $\mathbb{F}$ is exactly three, where two layers can be sufficient if we consider a smaller domain and/or codomain. We further show that the minimum width for representing all functions from $\mathbb{F}^d$ to $\mathbb{F}$ lies between $2d$ and $2d + 4$. In addition, if we restrict the domain to non-negative floats, it lies between d and $d + 4$, where it can be smaller for a smaller domain, even beyond d . Our results show that the existing results analyzed under exact arithmetic do not extend to the floating-point setup.

1. Introduction

Depth and width are fundamental parameters determining the expressive power of neural networks. Classical universal approximation results show that two-layer networks of unbounded width can approximate any continuous func-

tion on a compact domain if the activation function is non-polynomial (Cybenko, 1989; Hornik et al., 1989; Leshno et al., 1993; Pinkus, 1999). For the memorization problem that aims to fit an arbitrary number of finite input-output pairs, it is also known that two layers suffice (Baum, 1988; Huang & Babri, 1998; Huang, 2003). These results imply that the minimum depth required to enable universal approximation and memorization properties is exactly two.

As a dual question, the minimum width for universal approximation and memorization has also been studied. Several studies show that the minimum width to approximate any continuous function from a d -dimensional compact domain to $\mathbb{R}$ lies between d and $d + c$ for some constant c for various practical activation functions (Lu et al., 2017; Hanin & Selke, 2017; Johnson, 2019; Kidger & Lyons, 2020; Park et al., 2021; Cai, 2023; Kim et al., 2024; Shin et al., 2025). On the other hand, unlike the universal approximation results, width 2 is sufficient for memorization since any finite set in $\mathbb{R}^d$ can be easily projected into distinct scalar values (Park et al., 2021; Shin et al., 2025). However, these minimum width and depth results are derived under exact arithmetic. Hence, they cannot be directly applied to real neural network implementations on computers using a finite set of numbers and inexact machine operations with round-off errors, especially floating-point arithmetic (IEEE, 2019).

Recently, Park et al. (2024) initiated research on the expressive power of neural networks that have floating-point parameters and use floating-point operations for internal computations. They show that four-layer floating-point networks using ReLU or the binary step function as the activation function can represent all floating-point functions (i.e., from floating-point vectors to floating-point values). This result has also been extended to general activation functions (Hwang et al., 2025b) and floating-point interval arithmetic (Hwang et al., 2025a). Nevertheless, while existing results imply that the minimum depth of floating-point networks for representing floating-point functions is bounded by four, the tightness of this bound is unknown. Furthermore, there is no known bound for the minimum width.

In this work, we investigate the minimum depth and width of floating-point networks for representing floating-point functions when the activation function is ReLU. On the

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Table 1. A summary of known bounds on the minimum depth ($\ell_{\min}$) and width ($w_{\min}$) of ReLU networks for universal approximation and memorization under exact arithmetic, and floating-point function representation under floating-point arithmetic. Here, $\mathcal{X} \subset \mathbb{R}$ denotes a finite set, $\mathbb{F}$ denotes the set of finite floating-point numbers, $e_{\dagger} = 2^{q-2} + p + 4$ and $e_{\ddagger} = 2^{q-2} + p + 3$ where p denotes the bits for the mantissa and q denotes the bits for the exponent in floating-point formats, ω and Ω denote the smallest and largest positive finite floats, and $\mathcal{S}_{\mathbb{F}}$ denotes $\mathcal{S} \cap \mathbb{F}$. See Section 2 for the precise definitions.

Reference	Domain $\rightarrow$ Codomain	Upper/Lower Bounds
Pinkus (1999)	$[0, 1]^d \rightarrow \mathbb{R}$	$\ell_{\min} = 2$
Bubeck et al. (2020)	$\mathcal{X} \rightarrow \mathbb{R}$	$\ell_{\min} = 2$
Park et al. (2024)	$\mathbb{F}^d \rightarrow \mathbb{F}$	$\ell_{\min} \leq 4$
Ours (Theorem 1)	$\mathbb{F}^d \rightarrow \mathbb{F}$	$\ell_{\min} \leq 3$
Ours (Theorem 2)	$[-1, 1]_{\mathbb{F}} \rightarrow \mathbb{F}$	$\ell_{\min} \geq 3$
Ours (Theorem 3)	$[-1, 1]_{\mathbb{F}} \rightarrow [-1, 1]_{\mathbb{F}}$	$\ell_{\min} = 2$
Ours (Theorem 4)	$[0, \Omega]_{\mathbb{F}} \rightarrow \mathbb{F}$	$\ell_{\min} = 2$
Hanin & Sellke (2017)	$[0, 1]^d \rightarrow \mathbb{R}$	$w_{\min} = d + 1$
Park et al. (2021)	$\mathcal{X} \rightarrow \mathbb{R}$	$w_{\min} = 2$
Ours (Theorem 5)	$\mathbb{F}^d \rightarrow \mathbb{F}$	$w_{\min} \leq 2d + 4$
Ours (Theorem 6)	$(-2^{e_{\dagger}}, 2^{e_{\dagger}})_{\mathbb{F}}^d \rightarrow \mathbb{F}$	$w_{\min} \geq 2d$
Ours (Theorem 7)	$[0, \Omega]_{\mathbb{F}}^d \rightarrow \mathbb{F}$	$w_{\min} \leq d + 4$
Ours (Theorem 8)	$[0, 2^{e_{\ddagger}}]_{\mathbb{F}}^d \rightarrow \mathbb{F}$	$w_{\min} \geq d$
Ours (Theorem 9)	$\{0, \omega\}^d \rightarrow \mathbb{F}$	$w_{\min} \leq \lceil d/(p+1) \rceil + 4$

one hand, representing floating-point functions requires fitting finite input-output pairs, which is similar to the memorization problem. On the other hand, it is also related to universal approximation since any floating-point function can be interpolated by a continuous function. However, we show that the minimum width and depth for representing floating-point functions do not follow the existing universal approximation and memorization results. In particular, our contributions can be summarized as follows, where $\mathbb{F}$ denotes the set of finite floating-point numbers.

- We first show that three-layer floating-point networks can represent all functions from $\mathbb{F}^d$ to $\mathbb{F}$ (Theorem 1). This result improves upon existing four-layer constructions.
- This bound is tight even for $d = 1$: there exists a floating-point function that cannot be represented by a two-layer floating-point network (Theorem 2). Namely, representing floating-point functions requires strictly more layers than universal approximation and memorization under exact arithmetic, where two layers are sufficient.
- However, two layers can be sufficient for a smaller domain and/or codomain (Theorems 3 and 4). Such a dependency between the minimum depth and the domain/codomain does not arise under exact arithmetic since the minimum depth and width do not change under invertible affine transformations of the domain/codomain.
- We next show that to represent all functions from $\mathbb{F}^d$ to $\mathbb{F}$, width $2d + 4$ is sufficient (Theorem 5) and width $2d$ is necessary (Theorem 6). These results imply that representing floating-point functions requires strictly more

width than universal approximation and memorization under exact arithmetic, where width $d + 1$ and width 2 are sufficient for ReLU networks, respectively.

- Nevertheless, as in the minimum depth case, a smaller width can be sufficient for a smaller domain. If the inputs are non-negative floats, then width $d + 4$ is sufficient (Theorem 7) while width d is necessary (Theorem 8). This can be reduced further: if the inputs consist of zero and the smallest positive float, then width $\lceil d/(p+1) \rceil + 4$ suffices, where p is the number of bits in the mantissa.

2. Preliminaries

2.1. Notations

We use $\mathbb{R}$, $\mathbb{Z}$, $\mathbb{N}$, and $\overline{\mathbb{R}}$ to denote the set of real numbers, the set of integers, the set of positive integers, and $\mathbb{R} \cup \{\pm\infty\}$, respectively. For $n \in \mathbb{N}$, we define $[n] := \{1, \dots, n\}$. For $x \in \mathbb{R}$ and $\mathcal{S} \subset \mathbb{R}$, we use $\lceil x \rceil_{\mathcal{S}} := \inf\{z \in \mathcal{S} : z \geq x\}$. For $a, b \in \overline{\mathbb{R}}$ and a set $\mathcal{S}$, we define $(a, b) := \{x \in \mathbb{R} : a < x < b\}$, $[a, b] := \{x \in \mathbb{R} : a \leq x \leq b\}$, $(a, b)_{\mathcal{S}} := (a, b) \cap \mathcal{S}$, and $[a, b]_{\mathcal{S}} := [a, b] \cap \mathcal{S}$. We also define half-open intervals $[a, b)$, $(a, b]$, $[a, b)_{\mathcal{S}}$, and $(a, b]_{\mathcal{S}}$ similarly. For a set $\mathcal{X}$, $m, n \in \mathbb{N}$, $x \in \mathcal{X}^n$, $M \in \mathcal{X}^{n \times m}$, $i \in [n]$, and $j \in [m]$, we use x_i to denote the i -th coordinate of x , M_i to denote the i -th row of M , and M_{ij} to denote the element in the i -th row and the j -th column of M ; here, x is a column vector.

2.2. Floating-point numbers

In this paper, we follow the IEEE-754 standard for floating-point arithmetic (IEEE, 2019). For $p, q \in \mathbb{N}$, $\mathbb{F}$ denotes the

set of finite floating-point numbers defined as

$$\mathbb{F}_{p,q} := \{\pm(1+z) \times 2^e, \pm z \times 2^{\epsilon_{\min}} : z \in \mathbb{M}_p, e \in [\epsilon_{\min}, \epsilon_{\max}]\}$$

where $\mathbb{M}_p = \{(z-1)/2^p : z \in [2^p]\}$, $\epsilon_{\min} = -2^{q-1} + 2$, and $\epsilon_{\max} = 2^{q-1} - 1$. There are three additional non-finite floating-point numbers $-\infty, \infty, \text{NaN}$ where $\pm\infty$ are numbers satisfying $-\infty < x < \infty$ for all $x \in \mathbb{R}$, and NaN is a special symbol denoting *not a number*, which can be generated as an output of an invalid operation (e.g., $1/0$, $-\infty + \infty$). We use $\overline{\mathbb{F}}_{p,q}$ to denote the set of all floating-point numbers, i.e., $\overline{\mathbb{F}}_{p,q} = \mathbb{F}_{p,q} \cup \{\pm\infty, \text{NaN}\}$. Then, $\overline{\mathbb{F}}_{p,q}$ can be represented using $p+q+1$ bits: we use 1 bit for *sign*, p bits for *mantissa* (i.e., $x \in \mathbb{M}_p$), and q bits for *exponent*. Since we are only using $2^q - 2$ exponents in $\mathbb{F}_{p,q}$, we can represent $\pm\infty$ and NaN using the remaining exponent encodings. If p, q are clear from the context, we omit them and use $\overline{\mathbb{F}}$ and $\mathbb{F}$ to denote $\overline{\mathbb{F}}_{p,q}$ and $\mathbb{F}_{p,q}$, respectively.

We introduce the following additional notation. Given $p, q \in \mathbb{N}$, we use $\omega = 2^{\epsilon_{\min}-p}$ to denote the smallest positive float and $\Omega = (2 - 2^{-p}) \times 2^{\epsilon_{\max}}$ to denote the largest finite float. For $x \in \mathbb{F}$, we use x^+ (and x^-) to denote the smallest (and largest) floating-point number that is larger (and smaller) than x (e.g., $0^+ = \omega$ and $(-\Omega)^- = -\infty$).

Throughout this paper, we assume Condition 1 on p, q . We note that most practical floating-point formats satisfy this condition, e.g., double/single/half-precision formats (IEEE, 2019), bfloat16 (Google), and FP8 (E5M2, E4M3, Micikevicius et al. (2022)). See Table 2 in Appendix.

Condition 1. $2 \leq p \leq 2^{q-1} - 5$.

2.3. Floating-point operations

To define floating-point operations, we first introduce floating-point rounding $\lceil \cdot \rceil_{\mathbb{F}_{p,q}} : \mathbb{R} \cup \{\text{NaN}\} \rightarrow \overline{\mathbb{F}}_{p,q}$ as follows:

$$\lceil x \rceil_{\mathbb{F}_{p,q}} := \begin{cases} \arg \min_{y \in \mathbb{F}} |x - y| & \text{if } |x| < \Omega + 2^{\epsilon_{\max}-p-1}, \\ \infty & \text{if } x \geq \Omega + 2^{\epsilon_{\max}-p-1}, \\ -\infty & \text{if } x \leq -\Omega - 2^{\epsilon_{\max}-p-1}, \\ \text{NaN} & \text{if } x = \text{NaN}. \end{cases}$$

If there are two floating-point numbers closest to $x \in \mathbb{R}$, then we follow the *ties-to-even* rule (IEEE, 2019): choose $\lceil x \rceil_{\mathbb{F}_{p,q}}$ as the closest floating-point number that can be represented as $(1+z) \times 2^e$ or $z \times 2^{\epsilon_{\min}}$ for some $z \in \mathbb{M}_{p-1}$ and $e \in [\epsilon_{\min}, \epsilon_{\max}] \cap \mathbb{Z}$. If $\mathbb{F}_{p,q}$ is clear from the context, we use $\lceil \cdot \rceil$ to denote $\lceil \cdot \rceil_{\mathbb{F}_{p,q}}$.

For $x, y \in \mathbb{F}$, we define floating-point addition and multiplication as $x \oplus y = \lceil x + y \rceil$ and $x \otimes y = \lceil x \times y \rceil$. If x or y is NaN, then the outputs of these operations are also NaN. For $x = \pm\infty$ and $y \in \mathbb{F}$, $x \oplus y = y \oplus x = \pm\infty$. If $x = \pm\infty$ and $y = \mp\infty$, then $x \oplus y = \text{NaN}$. If $x = \pm\infty$

and $y > 0$ (or $y < 0$), then $x \otimes y = y \otimes x = \pm\infty$ (or $\mp\infty$). If $x = \pm\infty$ and $y = 0$, then $x \otimes y = y \otimes x = \text{NaN}$. We also define $x \ominus y = x \oplus (-y)$ where $-\text{NaN} = \text{NaN}$. We note that these operations follow the IEEE-754 standard.

Moreover, floating-point operations are *non-associative*; that is, $(x \odot y) \odot z \neq x \odot (y \odot z)$ in general for $\odot \in \{\otimes, \oplus\}$. Hence, specifying the order of operations is important. As in standard arithmetic, we compute $\otimes$ first and $\oplus$ next, i.e., $x \oplus y \otimes z = x \oplus (y \otimes z)$. Furthermore, we compute operations in a *left-associative* way: for $x_1, \dots, x_n \in \overline{\mathbb{F}}$,

$$x_1 \oplus \dots \oplus x_n = (\dots((x_1 \oplus x_2) \oplus x_3) \oplus \dots \oplus x_{n-1}) \oplus x_n.$$

We define $\bigoplus_{i=1}^n x_i = x_1 \oplus \dots \oplus x_n$ and also compute $\oplus$ and $\otimes$ in a left-associative way:

$$\bigoplus_{i=1}^n x_i \otimes \bigoplus_{j=1}^m y_j = x_1 \oplus \dots \oplus x_n \otimes y_1 \oplus \dots \oplus y_m. \quad (1)$$

We extend floating-point operations to matrices and vectors. For $n, m \in \mathbb{N}$, $M \in \overline{\mathbb{F}}^{m \times n}$, and $u, v \in \overline{\mathbb{F}}^n$, we define $M \otimes v \in \overline{\mathbb{F}}^m$ and $u \oplus v$ such that $(M \otimes v)_i = \bigoplus_{k=1}^n (M_{ik} \otimes v_k)$ and $(u \oplus v)_j = u_j \oplus v_j$ hold for all $i \in [m]$ and $j \in [n]$.

2.4. Floating-point neural networks

To define floating-point neural networks, we first define the activation function $\sigma : \overline{\mathbb{F}} \rightarrow \overline{\mathbb{F}}$ as $\sigma(x) = x$ if $x \geq 0$, $\sigma(x) = 0$ if $x < 0$, and $\sigma(\text{NaN}) = \text{NaN}$; σ follows the implementations of ReLU in PyTorch (Paszke et al., 2019) and TensorFlow (Abadi et al., 2016). We call $f : \overline{\mathbb{F}}^d \rightarrow \overline{\mathbb{F}}$ a *floating-point network* if there exist $L, d_0 = d, \dots, d_L \in \mathbb{N}$ and $\phi_l(x) = W_l \otimes x \oplus b_l$ for some $W_l \in \overline{\mathbb{F}}^{d_l \times d_{l-1}}$ and $b_l \in \overline{\mathbb{F}}^{d_l}$ for all $l \in [L]$ such that f is defined as

$$f(x) = \phi_L \circ \sigma \circ \dots \circ \sigma \circ \phi_1(x) \quad (2)$$

where σ is applied coordinate-wise. We define the *width* of a floating-point network f as $\max\{d_1, \dots, d_{L-1}\}$ and define the *depth* (or the number of layers) of f as L .

3. Minimum depth and width of FP networks for FP function representation

In this section, we present upper and lower bounds on the minimum depth and width of floating-point networks to represent floating-point functions, and we compare them with existing results derived under the exact arithmetic.

3.1. Minimum depth for FP function representation

We first show that three layers are sufficient to represent all functions from $\mathbb{F}^d$ to $\mathbb{F}$ in the following theorem. The proof of Theorem 1 is in Section 4.1.

Theorem 1. *For any $d \in \mathbb{N}$ and $f^* : \mathbb{F}^d \rightarrow \mathbb{F}$, there exists a three-layer floating-point network $f : \mathbb{F}^d \rightarrow \mathbb{F}$ such that $f(x) = f^*(x)$ for all $x \in \mathbb{F}^d$.*

Theorem 1 improves the number of layers from the existing four-layer network constructions in (Park et al., 2024; Hwang et al., 2025b) to three layers. We further show that Theorem 1 is tight with respect to the number of layers: two layers are generally insufficient, as stated in the following theorem. The proof of Theorem 2 is in Section 4.3.

Theorem 2. *Suppose $p \leq 2^{q-1} - 7$. Then, there exists $f^* : [-1, 1]_{\mathbb{F}} \rightarrow \mathbb{F}$ such that for any two-layer floating-point network $f : \mathbb{F} \rightarrow \mathbb{F}$, $f(x) \neq f^*(x)$ for some $x \in [-1, 1]_{\mathbb{F}}$.*

Classical universal approximation and memorization results show that two-layer networks can approximate any continuous function on a compact domain (Pinkus, 1999) and can fit any finite input-output pairs (Bubeck et al., 2020) for various activation functions. Unlike these results, Theorems 1 and 2 implies that three layers are necessary and sufficient for floating-point (ReLU) networks to represent all floating-point functions from $\mathbb{F}^d \rightarrow \mathbb{F}$. As a floating-point function can be interpolated by continuous functions on a compact domain or regarded as finite input/output pairs, our results show that results under exact arithmetic do not generalize to floating-point arithmetic. Note that the condition on p and q in Theorem 2 is satisfied in single/double-precision formats, bfloat16, and FP8 (E5M2); see Table 2 in Appendix.

We prove Theorem 1 by explicitly constructing indicator functions $\mathbb{1}_{x=z}$ ¹ implemented by a two-layer floating-point network for all $z \in \mathbb{F}^d$. We then represent the target function as a sum of weights multiplied by indicator functions:

$$f(x) = \bigoplus_{i=1}^{|\mathbb{F}^d|} (f^*(z_i) \otimes \mathbb{1}_{x=z_i})$$

where $z_1, \dots, z_{|\mathbb{F}^d|}$ denote all elements in $\mathbb{F}^d$. Then, f can be implemented by a three-layer floating-point network.

To prove Theorem 2, we show that for any two-layer network f , if $f(0) = -\Omega$ and $f(\omega) = f(-\omega) = \Omega$, then $f(1) \notin \mathbb{F}$ or $f(-1) \notin \mathbb{F}$. Specifically, we prove that to construct the highly oscillating parts $f(0) = -\Omega$ and $f(\omega) = f(-\omega) = \Omega$, an overflow must occur during the intermediate computation of $f(1)$ or $f(-1)$. To enforce the overflow, we use not only the wide range $\mathbb{F}$, but also properties of Ω and both the positive/negative parts of inputs.

The counterexample function in the proof of Theorem 2 has a wide range $\mathbb{F}$ and contains both the positive and negative parts in the domain. We further show in Theorems 3 and 4 that these properties are necessary to demonstrate the insufficiency of two layers. We present the proofs of Theorems 3 and 4 in Appendices C.1 and C.2.

¹ $\mathbb{1}_{x=z} = 1$ if $x = z$ and $\mathbb{1}_{x=z} = 0$ if $x \neq z$

Theorem 3. *For any $f^* : [-1, 1]_{\mathbb{F}} \rightarrow [-1, 1]_{\mathbb{F}}$, there exists a two-layer floating-point network $f : \mathbb{F} \rightarrow \mathbb{F}$ such that $f(x) = f^*(x)$ for all $x \in [-1, 1]_{\mathbb{F}}$.*

Theorem 4. *For any $f^* : [0, \infty)_{\mathbb{F}} \rightarrow \mathbb{F}$, there exists a two-layer floating-point network $f : \mathbb{F} \rightarrow \mathbb{F}$ such that $f(x) = f^*(x)$ for all $x \in [0, \infty)_{\mathbb{F}}$.*

Theorem 3 shows that if we reduce the range from $\mathbb{F}$ to $[-1, 1]_{\mathbb{F}}$ (compared to Theorem 2), then two layers are sufficient. In particular, we can bypass the overflow that arises in the proof of Theorem 2 by considering a smaller range. We note that two layers are necessary since all one-layer floating-point networks are coordinate-wise monotone. Theorem 4 shows that if the domain consists of non-negative numbers, then two layers are sufficient. That is, if we discard negative (or symmetrically positive) inputs, then we can bypass the overflow in the proof of Theorem 2.

Theorems 3 and 4 show that considering a different domain or codomain can reduce the required number of layers to represent floating-point functions. Such an observation does not hold in the classical setting: two layers are sufficient for the classical universal approximation under exact arithmetic, as long as the domain is compact and the target function is continuous, i.e., a compact codomain (Pinkus, 1999). This is because, under exact arithmetic, we can easily transform a unit cube (e.g., domain or codomain) to any other cube via an invertible affine transformation without any information loss. However, under floating-point arithmetic, this is not the case due to overflow and round-off errors.

3.2. Minimum width for FP function representation

In this section, we present our upper and lower bounds on the minimum width of floating-point networks for representing floating-point functions. We first show that width $2d + 4$ is sufficient to represent all functions from $\mathbb{F}^d$ to $\mathbb{F}$ in the following theorem, whose proof is in Appendix C.3.

Theorem 5. *For any $d \in \mathbb{N}$ and $f^* : \mathbb{F}^d \rightarrow \mathbb{F}$, there exists a floating-point network $f : \mathbb{F}^d \rightarrow \mathbb{F}$ of width $2d + 4$ such that $f(x) = f^*(x)$ for all $x \in \mathbb{F}^d$.*

In the proof of Theorem 5, we use width $2d$ to preserve the d -dimensional input information (one for the positive value and one for the negative value for each dimension), width 2 to compute intermediate values, and width 2 to store the intermediate/final output.

Using the bit-counting argument, one might expect that width $2d$ is not necessary: the number of possible inputs is $|\mathbb{F}^d| = (2^{p+1}(2^q - 1) - 1)^d \approx 2^{d(p+q+1)}$, but the number of possible outputs after applying the first affine layer and $\lceil \text{ReLU} \rceil$ is $|[0, \infty)_{\mathbb{F}}|^{2d} = (2^p(2^q - 1))^{2d} \approx 2^{2d(p+q)}$. Here, we exclude ∞ , NaN from the range since they would destroy the final output (see Lemma 26 in Appendix). That is, even for a small value of d , we have $|\mathbb{F}^d| \leq |[0, \infty)_{\mathbb{F}}|^{2d-1}$. Then,

can we represent all functions from $\mathbb{F}^d$ to $\mathbb{F}$ using a width strictly smaller than $2d$, or even close to d ? In the following theorem, we prove that the answer is *no* in general, even for a smaller domain. See Appendix C.4 for its proof.

Theorem 6. *Let $e^* = (\epsilon_{\max} + 2p + 9)/2$ and suppose that $e^* \leq \epsilon_{\max} + 1$. For any $d \in \mathbb{N}$, there exists $f^* : (-2^{e^*}, 2^{e^*})_{\mathbb{F}}^d \rightarrow \mathbb{F}$ such that for any floating-point network f of width $2d - 1$, $f(x) \neq f^*(x)$ for some $x \in (-2^{e^*}, 2^{e^*})_{\mathbb{F}}^d$.*

Theorem 6 states that width $2d$ is necessary to represent all functions from $(-2^{e^*}, 2^{e^*})_{\mathbb{F}}^d$ to $\mathbb{F}$ when $e^* = (\epsilon_{\max} + 2p + 9)/2 \leq \epsilon_{\max} + 1$. Here, the condition on e^* is satisfied for most floating-point formats used in practice: e.g., single/double-precision formats, bfloat16, and FP8 (E5M2). To prove Theorem 6, we choose a finite set of input pairs (say $\mathcal{X}$). Then, we show that for any floating-point network of width $\leq 2d - 1$, at least one pair of inputs in $\mathcal{X}$ should have the same output by exploiting the round-off error. See Appendix C.4 for more details.

Compared to existing results under exact arithmetic, the lower bound $2d$ on the minimum width is strictly larger. Under exact arithmetic, recent universal approximation results find that width $d + c$ suffices for approximating arbitrary continuous functions from a compact domain to $\mathbb{R}$ for some constant c , for various activation functions (Kidger & Lyons, 2020; Shin et al., 2025). In particular, width $d + 1$ is necessary and sufficient for ReLU networks (Hanin & Sellke, 2017; Park et al., 2024; Kim et al., 2024). This is because width d is sufficient for preserving the input information; for the input domain $[-k, k]^d$, consider $x \mapsto \text{ReLU}(x + (k, \dots, k))$. However, such a transformation is impossible under floating-point arithmetic since non-trivial addition is not injective in general due to round-off error. Furthermore, to fit finite input/output pairs, width 2 is sufficient for real ReLU networks (Park et al., 2021). This can be done by projecting a finite number of d -dimensional inputs to distinct scalar values, which is also impossible under floating-point arithmetic.

We next show that if the domain consists of non-negative numbers only, then width $d + 4$ is sufficient (Theorem 7) and width d is necessary (Theorem 8). The proofs of Theorems 7 and 8 are presented in Sections 4.2 and 4.4.

Theorem 7. *For any $d \in \mathbb{N}$ and $f^* : [0, \infty)_{\mathbb{F}}^d \rightarrow \mathbb{F}$, there exists a floating-point network $f : \mathbb{F}^d \rightarrow \mathbb{F}$ of width $d + 4$ such that $f(x) = f^*(x)$ for all $x \in [0, \infty)_{\mathbb{F}}^d$.*

Theorem 8. *Let $e^* = (\epsilon_{\max} + 2p + 7)/2$ and suppose that $e^* \leq \epsilon_{\max} + 1$. For any $d \in \mathbb{N}$, there exists $f^* : [0, 2^{e^*})_{\mathbb{F}}^d \rightarrow \mathbb{F}$ such that for any floating-point network f of width $d - 1$, $f(x) \neq f^*(x)$ for some $x \in [0, 2^{e^*})_{\mathbb{F}}^d$.*

Compared to Theorem 5, we reduce width d in Theorem 7. This is possible because preserving the d -dimensional input information can be done using width d : e.g., consider $x \mapsto$

$\sigma(x)$. As in our depth analysis in Section 3.1, Theorems 6 and 7 imply that the required width to represent floating-point functions depends on the domain. We note that such an observation cannot be made under exact arithmetic for both universal approximation and memorization setups.

We finally show that the required width can be smaller than d when the domain is tiny. This result contrasts with the universal approximation results under exact arithmetic, which generally require a width of $\geq d$. Specifically, we prove that to represent all functions from $[0, \omega]_{\mathbb{F}}^d$ to $\mathbb{F}$, width $\lceil d/(p + 1) \rceil_{\mathbb{Z}} + 4$ suffices. Here, note that $[0, \omega]_{\mathbb{F}} = \{0, \omega\}$. We present the proof of Theorem 9 in Appendix C.5.

Theorem 9. *For any $d \in \mathbb{N}$ and $f^* : [0, \omega]_{\mathbb{F}}^d \rightarrow \mathbb{F}$, there exists a floating-point network $f : \mathbb{F}^d \rightarrow \mathbb{F}$ of width $\lceil d/(p + 1) \rceil_{\mathbb{Z}} + 4$ such that $f(x) = f^*(x)$ for all $x \in [0, \omega]_{\mathbb{F}}^d$.*

4. Proofs

In this section, we prove Theorems 1, 2, 7, and 8. To better present our proofs, we first prove the upper bounds on the minimum depth and width (Theorems 1 and 7) in Sections 4.1 and 4.2, and prove the corresponding lower bounds (Theorems 2 and 8) later in Sections 4.3 and 4.4.

4.1. Proof of Theorem 1

We first introduce the following technical lemma. The proof of Lemma 10 is deferred to Appendix C.6.

Lemma 10. *For any $z \in \mathbb{F}$ and $s \in \{-1, 1\}$, there exist $u, w, b \in \mathbb{F}$ such that for each $x \in \mathbb{F}$,*

- $f(x) = u \otimes \sigma(w \otimes x \oplus b) = 0$ if $s \times x \geq s \times z$ and
- $f(x) \in [-\Omega, -\omega]_{\mathbb{F}}$ otherwise.

Let $m = |\mathbb{F}|^d$, $\{z_1, \dots, z_m\} = \mathbb{F}^d$, and $z_i = (z_{i,1}, \dots, z_{i,d})$ for all $i \in [m]$. For each $i \in [m]$, $j \in [d]$, and $k \in [2]$, we choose $u_{i,j,k}, w_{i,j,k}, b_{i,j,k} \in \mathbb{F}$ using Lemma 10 such that

$$\begin{aligned} u_{i,j,1} \otimes \sigma(w_{i,j,1} \otimes y \oplus b_{i,j,1}) &= 0 & \text{if } y \geq z_{i,j}, \\ -\Omega \leq u_{i,j,1} \otimes \sigma(w_{i,j,1} \otimes y \oplus b_{i,j,1}) &\leq -\omega & \text{if } y < z_{i,j}, \\ u_{i,j,2} \otimes \sigma(w_{i,j,2} \otimes y \oplus b_{i,j,2}) &= 0 & \text{if } y \leq z_{i,j}, \\ -\Omega \leq u_{i,j,2} \otimes \sigma(w_{i,j,2} \otimes y \oplus b_{i,j,2}) &\leq -\omega & \text{if } y > z_{i,j}, \end{aligned}$$

for all $y \in \mathbb{F}$, i.e., $\sigma(w_{i,j,k} \otimes y \oplus b_{i,j,k}) \in \mathbb{F}$ for all $k \in [2]$ and $y \in \mathbb{F}$. For each $i \in [m]$, let $h_i : \mathbb{F} \rightarrow \mathbb{F}$ be a two-layer floating-point network defined as

$$\begin{aligned} h_i(x) &= \sigma(\omega) \oplus \bigoplus_{j=1}^d (u_{i,j,1} \otimes \sigma(w_{i,j,1} \otimes x_j \oplus b_{i,j,1})) \\ &\quad \oplus \bigoplus_{j=1}^d (u_{i,j,2} \otimes \sigma(w_{i,j,2} \otimes x_j \oplus b_{i,j,2})). \end{aligned} \quad (3)$$

$$\begin{aligned}
 & \begin{bmatrix} x \\ 0 \\ 0 \\ \alpha_{i-1,x} \\ \beta_{i-1,x} \end{bmatrix} \rightarrow \begin{bmatrix} x \\ \sigma(\hat{w}_{i,0} \otimes x_{\iota_0} \oplus \hat{b}_{i,0}) \\ 0 \\ \alpha_{i-1,x} \\ \beta_{i-1,x} \end{bmatrix} \rightarrow \begin{bmatrix} x \\ \sigma(\hat{w}_{i,1} \otimes x_{\iota_1} \oplus \hat{b}_{i,1}) \\ \gamma_{i,0} = \sigma(\hat{u}_{i,0} \otimes \sigma(\hat{w}_{i,0} \otimes x_{\iota_0} \oplus \hat{b}_{i,0})) \\ \alpha_{i-1,x} \\ \beta_{i-1,x} \end{bmatrix} \rightarrow \begin{bmatrix} x \\ \sigma(\hat{w}_{i,2} \otimes x_{\iota_2} \oplus \hat{b}_{i,3}) \\ \gamma_{i,1} = \sigma(\gamma_{i,0} \oplus \hat{u}_{i,1} \otimes \sigma(\hat{w}_{i,1} \otimes x_{\iota_1} \oplus \hat{b}_{i,1})) \\ \alpha_{i-1,x} \\ \beta_{i-1,x} \end{bmatrix} \\
 & \cdots \rightarrow \begin{bmatrix} x \\ \sigma(\hat{w}_{i,j+1} \otimes x_{\iota_{j+1}} \oplus \hat{b}_{i,j+1}) \\ \gamma_{i,j} = \sigma(\gamma_{i,j-1} \oplus \hat{u}_{i,j} \otimes \sigma(\hat{w}_{i,j} \otimes x_{\iota_j} \oplus \hat{b}_{i,j})) \\ \alpha_{i-1,x} \\ \beta_{i-1,x} \end{bmatrix} \cdots \rightarrow \begin{bmatrix} x \\ 0 \\ \mathbb{1}_{x=z_i} \\ \alpha_{i-1,x} \\ \beta_{i-1,x} \end{bmatrix} \rightarrow \begin{bmatrix} x \\ 0 \\ 0 \\ \sigma(\alpha_{i-1,x} \oplus \max\{f^*(z_i), 0\} \otimes \mathbb{1}_{x=z_i}) \\ \sigma(\beta_{i-1,x} \oplus \max\{-f^*(z_i), 0\} \otimes \mathbb{1}_{x=z_i}) \end{bmatrix} = \begin{bmatrix} x \\ 0 \\ 0 \\ \alpha_{i,x} \\ \beta_{i,x} \end{bmatrix}
 \end{aligned}$$

 Figure 1. Illustration of the construction of f_i in the proof of Theorem 7.

We note that the summation in the above equation is computed in a left-associative way (see Eq. (1)). Then, by the definitions of $u_{i,j,k}$, $w_{i,j,k}$, and $b_{i,j,k}$, we have $h_i(x) = \omega$ if $x = z_i$ and $h_i(x) \leq 0$ if $x \neq z_i$.

We next use the following lemma. The proof of Lemma 11 is deferred to Appendix C.7.

Lemma 11. *There exist $n \in \mathbb{N}$ and $\alpha_1, \dots, \alpha_n \in \mathbb{F}$ such that for any $j \in [n]$ and a floating-point number $x \leq \omega$,*

- $x \oplus \bigoplus_{i=1}^n \alpha_i = 1$ and $x \oplus \bigoplus_{i=1}^j \alpha_i > 0$ if $x = \omega$,
- $x \oplus \bigoplus_{i=1}^n \alpha_i \leq 0$ if $x \leq 0$, and
- $x \oplus \bigoplus_{i=1}^j \alpha_i \geq 0$ if $x = 0$.

Let $\alpha_1, \dots, \alpha_n$ be the floating-point numbers in Lemma 11, and $g(x) : \mathbb{F}^d \rightarrow \mathbb{F}^m$ be a two-layer network such that for each $i \in [m]$ and $x = (x_1, \dots, x_d) \in \mathbb{F}^d$,

$$g(x)_i = h_i(x) \oplus \bigoplus_{j=1}^n (\alpha_j \otimes \sigma(1)). \quad (4)$$

Then, by the definition of g , one can observe that $g(x)_i = 1$ if $x = z_i$ and $-\infty \leq g(x)_i \leq 0$ if $x \in \mathbb{F}^d \setminus \{z_i\}$. We design a target three-layer network f as

$$f(x) = (f^*(z_1), \dots, f^*(z_m))^T \otimes (\sigma \circ g(x)). \quad (5)$$

Then, one can observe that $f = f^*$ on $\mathbb{F}^d$, which completes the proof of Theorem 1.

4.2. Proof of Theorem 7

In this proof, we convert the shallow and wide network construction in the proof of Theorem 1 into a deep and narrow architecture. Let $\{z_1, \dots, z_{\hat{m}}\} = [0, \infty)_{\mathbb{F}}^d$. Recall Eqs. (3)–(5), and define $\hat{u}_{i,j}, \hat{w}_{i,j}, \hat{b}_{i,j} \in \mathbb{F}$ and $\iota_j \in [d]$ for all $j \in [2d+n] \cup \{0\}$ as follows:

$$\hat{u}_{i,j}, \hat{w}_{i,j}, \hat{b}_{i,j}, \iota_j = \begin{cases} 1, 0, \omega, 1 & \text{if } j = 0, \\ u_{i,k,1}, w_{i,k,1}, b_{i,k,1}, k & \text{if } j = k, k \in [d], \\ u_{i,k,2}, w_{i,k,2}, b_{i,k,2}, k & \text{if } j = d+k, k \in [d], \\ \alpha_k, 0, 1, 1 & \text{if } j = 2d+k, k \in [n]. \end{cases}$$

Then, by the definitions of $u_{i,j}, w_{i,j}, b_{i,j}$, and α_j , we have $\sigma(\bigoplus_{j=0}^{2d+n} (\hat{u}_{i,j} \otimes \sigma(\hat{w}_{i,j} \otimes x_{\iota_j} \oplus \hat{b}_{i,j}))) = \mathbb{1}_{x=z_i}$ (the inner sum is equal to $g(x)_i$ in Eq. (4)). We will transform this summation into a deep and narrow architecture.

For $i \in [\hat{m}]$ and $j \in [2d+n] \cup \{0\}$, define $\gamma_{i,-1} = 0$ and

$$\gamma_{i,j} = \sigma(\gamma_{i,j-1} \oplus \hat{u}_{i,j} \otimes \sigma(\hat{w}_{i,j} \otimes x_{\iota_j} \oplus \hat{b}_{i,j})).$$

Then, by the properties of α_j in Lemma 11 and definitions of $u_{i,j,k}, w_{i,j,k}, b_{i,j,k}$ in Section 4.1, we also have $\gamma_{i,2d+n} = \mathbb{1}_{x=z_i}$. For each $i \in [\hat{m}]$ and $x \in [0, \infty)_{\mathbb{F}}^d$, let

$$\begin{aligned}
 \alpha_{i,x} &= \bigoplus_{k=1}^i \max\{f^*(z_k), 0\} \otimes \mathbb{1}_{x=z_k}, \\
 \beta_{i,x} &= \bigoplus_{k=1}^i \max\{-f^*(z_k), 0\} \otimes \mathbb{1}_{x=z_k}.
 \end{aligned}$$

Then, one can observe that $\alpha_{\hat{m},x} \oplus (-\beta_{\hat{m},x}) = f^*(x)$.

We are now ready to construct the target network f . To this end, for each $i \in [\hat{m}]$, we construct $f_i : [0, \infty)_{\mathbb{F}}^{d+4} \rightarrow [0, \infty)_{\mathbb{F}}^{d+4}$ of the following form:

$$f_i(z) = \sigma \circ \phi_{i,L_i} \circ \sigma \circ \cdots \circ \sigma \circ \phi_{i,1}(z)$$

for some $L_i \in \mathbb{N}$ and floating-point affine transformations $\phi_{i,k} : \mathbb{F}^{d+4} \rightarrow \mathbb{F}^{d+4}$. In particular, we design f_i so that

$$f_i(x, 0, 0, \alpha_{i-1,x}, \beta_{i-1,x}) = (x, 0, 0, \alpha_{i,x}, \beta_{i,x}),$$

by using d neurons to store x , one neuron to compute $\sigma(\hat{w}_{i,j} \otimes x_{\iota_j} \oplus \hat{b}_{i,j})$, one neuron to compute $\gamma_{i,j}$, and two neurons to store $\alpha_{i-1,x}, \beta_{i-1,x}$ and compute $\alpha_{i,x}, \beta_{i,x}$; see Figure 1 for an illustration. Using f_i , we construct the target network f as follows:

$$f(x) = (0, \dots, 0, 1, -1)^T \otimes (f_{\hat{m}} \circ \cdots \circ f_1 \circ \sigma \circ \rho(x))$$

where ρ is a floating-point affine transformation that maps $x \in [0, \infty)_{\mathbb{F}}^d$ to $(x, 0, 0, 0, 0)$. Then, $f(x) = \alpha_{\hat{m},x} \oplus (-\beta_{\hat{m},x}) = f^*(x)$ for all $x \in [0, \infty)_{\mathbb{F}}^d$, which completes the proof.

4.3. Proof of Theorem 2

We argue by contradiction. Let f^* be a function from $[-1, 1]_{\mathbb{F}}$ to $\mathbb{F}$ such that

$$f^*(-\omega) = f^*(\omega) = \Omega, \text{ and } f^*(0) = -\Omega.$$

Suppose that there exists a two-layer floating-point network

$$f(x) = \left(\bigoplus_{i=1}^n (w_{2,i} \otimes \sigma(w_{1,i} \otimes x \oplus b_{1,i})) \right) \oplus b$$

such that $f = f^*$ on $[-1, 1]_{\mathbb{F}}$. Let

$$f_j(x) = \bigoplus_{i=1}^j (w_{2,i} \otimes \sigma(w_{1,i} \otimes x \oplus b_{1,i})),$$

$$g_j(x) = w_{2,j} \otimes \sigma(w_{1,j} \otimes x \oplus b_{1,j}),$$

for all $j \in [n] \cup \{0\}$. Then, to obtain $f = f^*$, one can observe that $f_j(x)$ and $g_j(x)$ must be finite for all $x \in [-1, 1]_{\mathbb{F}}$. In this proof, we will show the following: if $f(-\omega) = f(\omega) = \Omega$ and $f(0) = -\Omega$, then $g_k(1) = \infty$ or $g_k(-1) = \infty$ for some $k \in [n]$, i.e., $f(1) \notin \mathbb{F}$ or $f(-1) \notin \mathbb{F}$. Namely, there is no two-layer floating-point network that represents f^* on $[-1, 1]_{\mathbb{F}}$, which will complete the proof.

We first claim that the target values at $-\omega$, 0 , and ω must be constructed in f_n before adding the bias term b .

Claim 12. $f_n(-\omega) = f_n(\omega) = \Omega$ and $f_n(0) = -\Omega$.

Proof. Suppose that the claim is false. Then, b must satisfy $b > 2^{\epsilon_{\max}-p-1}$ or $b < -2^{\epsilon_{\max}-p-1}$ to have $f_n(-\omega) \oplus b = f_n(\omega) \oplus b = \Omega$ and $f_n(0) \oplus b = -\Omega$. However, if $b > 2^{\epsilon_{\max}-p-1}$, then there is no $z \in \mathbb{F}$ such that $z \oplus b = -\Omega$, i.e., $f_n(0) \oplus b$ cannot be $-\Omega$. Likewise, if $b < -2^{\epsilon_{\max}-p-1}$, then $f_n(-\omega) \oplus b$ and $f_n(\omega) \oplus b$ cannot be Ω . Hence, it must hold that $f_n(-\omega) = f_n(\omega) = \Omega$ and $f_n(0) = -\Omega$. $\square$

Given Claim 12, let $k \in [n]$ be the smallest number such that $f_j(-\omega) = f_j(\omega) = \Omega$ and $f_j(0) = -\Omega$ for all $j \geq k$. Then, we claim the following statement.

Claim 13. One of the following holds:

- $f_{k-1}(-\omega) = \Omega$, $f_{k-1}(0) = -\Omega$, and $f_{k-1}(\omega) < \Omega$.
- $f_{k-1}(-\omega) < \Omega$, $f_{k-1}(0) = -\Omega$, and $f_{k-1}(\omega) = \Omega$.

Proof. By the definition of k , we have $f_{k-1}(-\omega) < \Omega$, $f_{k-1}(0) > -\Omega$, or $f_{k-1}(\omega) < \Omega$. First, consider the case that $f_{k-1}(0) > -\Omega$. Then, to have $f_k(0) = -\Omega$, it must hold that $g_k(0) < -2^{\epsilon_{\max}-p-1}$. In this case, since g_k is monotone, we have $g_k(-\omega) < -2^{\epsilon_{\max}-p-1}$ or $g_k(\omega) < -2^{\epsilon_{\max}-p-1}$, which implies $f_k(-\omega) < \Omega$ or $f_k(\omega) < \Omega$. Hence, we cannot have $f_k(-\omega) = f_k(\omega) = \Omega$.

Now, consider the remaining case that $f_{k-1}(0) = -\Omega$ but $f_{k-1}(-\omega), f_{k-1}(\omega) < \Omega$. Then, $g_k(-\omega), g_k(\omega) > 2^{\epsilon_{\max}-p-1}$, which implies $g_k(0) > 2^{\epsilon_{\max}-p-1}$ by the monotonicity of g_k . In this case, we must have $f_k(0) > -\Omega$. Hence, the claim is valid. $\square$

Given Claims 12 and 13, without loss of generality, suppose that $f_{k-1}(-\omega) = \Omega$ and $f_{k-1}(0) = -\Omega$ but $f_{k-1}(\omega) < \Omega$. We next claim that $w_{2,k}, w_{1,k} > 0$ and $b_{1,k} \geq 0$ where $w_{2,k}, w_{1,k}, b_{1,k}$ are in the definition of g_k .

Claim 14. $w_{2,k}, w_{1,k} > 0$ and $b_{1,k} \geq 0$.

Proof. Since $f_k(-\omega) = f_k(\omega) = \Omega$ and $f_k(0) = -\Omega$, one can observe that $g_k(\omega) > 2^{\epsilon_{\max}-p-1}$ and $g_k(0) < 2^{\epsilon_{\max}-p-1}$. First, consider the case $w_{2,k} \leq 0$. In this case, since $\sigma(z) \geq 0$ for all $z \in \mathbb{F}$, it holds that $g_k(\omega) \leq 0$. Hence, we must have $w_{2,k} > 0$. If $w_{2,k} > 0$ and $w_{1,k} \leq 0$, then $g_k(\omega) \leq g_k(0) < 2^{\epsilon_{\max}-p-1}$, which implies that we must have $w_{1,k}, w_{2,k} > 0$.

Now, we assume $w_{1,k}, w_{2,k} > 0$ and consider the only remaining case that $b_{1,k} < 0$. In this case, since

$$w_{2,k} \otimes (w_{1,k} \otimes \omega) \geq g_k(\omega) > 2^{\epsilon_{\max}-p-1},$$

it holds that $w_{1,k}, w_{2,k} \geq 2^p$ (otherwise one of $w_{1,k}, w_{2,k}$ should exceed Ω). This implies that

$$\begin{aligned} \lceil w_{2,k} \times w_{1,k} \times \omega \rceil &= w_{2,k} \otimes \sigma(w_{1,k} \otimes \omega) \\ &\geq w_{2,k} \otimes \sigma(w_{1,k} \otimes \omega \oplus b_{1,k}) = g_k(\omega) > 2^{\epsilon_{\max}-p-1}, \end{aligned}$$

where the first equality is from $w_{1,k}, w_{2,k} \geq 2^p$ and ω is a power of two. Furthermore, since $g_k(\omega) \neq g_k(0)$, we must have $w_{1,k} \otimes \omega > -b_{1,k}$. Since $1 \geq 2^{2p+3} \times \omega$, it holds that

$$\begin{aligned} g_k(1) &= w_{2,k} \otimes \sigma(w_{1,k} \otimes 1 \oplus b_{1,k}) = w_{2,k} \otimes \sigma(w_{1,k} \otimes 1) \\ &= \lceil w_{1,k} \times w_{2,k} \rceil \geq 2^{2p+3} \lceil w_{1,k} \times w_{2,k} \times \omega \rceil \geq 2^{\epsilon_{\max}+p+2}. \end{aligned}$$

This implies that $g_k(1) = \infty$. Hence, the claim is valid. $\square$

Finally, we claim that given Claims 12–14, $f_k(-\omega) = f_k(\omega) = \Omega$, and $f_k(0) = -\Omega$, $g_k(1)$ must be infinite.

Claim 15. $g_k(1) = \infty$.

Proof. If $w_{1,k} \in (2^{m-1}, 2^m]_{\mathbb{F}}$ for some $m \geq 0$ (otherwise, $w_{1,k} \otimes \omega = 0 = w_{1,k} \otimes 0$), then $w_{1,k} \otimes \omega \in [0, 2^{\epsilon_{\min}-p+m}]_{\mathbb{F}}$. Hence, to have $g_k(\omega) \neq g_k(0)$, i.e.,

$$w_{1,k} \otimes \omega \oplus b_{1,k} \neq w_{1,k} \otimes 0 \oplus b_{1,k},$$

it must hold that $b_{1,k} < 2^{\epsilon_{\min}+m+2}$. This implies that $w_{1,k} \otimes \omega \oplus b_{1,k} \leq 2^{\epsilon_{\min}+m+2}$. Furthermore, to have $g_k(\omega) > 2^{\epsilon_{\max}-p-1}$, we need $w_{2,k} > 2^{\epsilon_{\max}-\epsilon_{\min}-p-m-3}$. However, since $p \leq 2^{q-1} - 7$, we have

$$w_{1,k} \times w_{2,k} > 2^{\epsilon_{\max}-\epsilon_{\min}-p-4} \geq 2^{\epsilon_{\max}+1},$$

i.e., $g_k(1) \geq w_{2,k} \otimes (w_{1,k} \otimes 1) = \infty$. $\square$

We note that $g_k(-1) = \infty$ if we assumed $f_{k-1}(\omega) = \Omega$ and $f_{k-1}(0) = -\Omega$ after Claim 13. Since $g_k(1) = \infty$, it holds that $f(1) \notin \mathbb{F}$. Thus, there is no two-layer network f satisfying $f = f^*$ on $[-1, 1]_{\mathbb{F}}$, which completes the proof of Theorem 2.

4.4. Proof of Theorem 8

In this proof, we will explicitly construct $f^* : [0, 2^{e^*}]_{\mathbb{F}}^d \rightarrow \mathbb{F}$ that cannot be represented by any floating-point network of width strictly smaller than d . Without loss of generality, we assume $d \geq 2$ here. For each $i \in [d]$, we define $x_i = (x_{i1}, \dots, x_{id})$, $x'_i = (x'_{i1}, \dots, x'_{id}) \in [0, 2^{e^*}]_{\mathbb{F}}^d$ such that for any $j \neq i$,

$$x_{ij} = x'_{ij} = (2^{e^*})^- \text{ and } 0 = x_{ii} \neq x'_{ii} = \omega = 2^{\epsilon_{\min} - p}.$$

Here, although $2^{\epsilon_{\max} + 1} \notin \mathbb{F}$, we let $(2^{\epsilon_{\max} + 1})^- = (2 - 2^{-p})2^{\epsilon_{\max}} = \Omega$ for the case that $e^* = \epsilon_{\max} + 1$. Choose $f^* : [0, 2^{e^*}]_{\mathbb{F}}^d \rightarrow \mathbb{F}$ such that

$$f^*(x_i) \neq f^*(x'_i) \text{ for all } i \in [d].$$

We now show that for any floating-point network f of width $d - 1$, $f(x_i) = f(x'_i)$, $f(x_i) = \text{NaN}$, or $f(x'_i) = \text{NaN}$ for some $i \in [d]$, and hence, it cannot represent f^* .

Let $z \mapsto W \otimes z \oplus b$ be the floating-point affine transformation in the first layer of f for some $d' \leq d - 1$, $W \in \mathbb{F}^{d' \times d}$, and $b \in \mathbb{F}^{d'}$. Let $W_i = (W_{i1}, \dots, W_{id})^\top$ be the i -th row of W for all $i \in [d']$. We will demonstrate the following statement: if $W_t \otimes x_i \neq W_t \otimes x'_i$ for some $i \in [d]$ and $t \in [d']$, then one of the following holds:

- (a) $W_t \otimes x_j = W_t \otimes x'_j$ for all $j \neq i$.
- (b) $W_t \otimes x_j = \text{NaN}$ or $W_t \otimes x'_j = \text{NaN}$ for some $j \in [d]$.

Suppose that this statement is true. Then, since $W_k \otimes x_j = \text{NaN}$ implies $f(x_j) = \text{NaN} \neq f^*(x_j) \in \mathbb{F}$, we may assume that (b) does not occur. If we assume (a), then by the pigeonhole principle and since $d' \leq d - 1$, we have $f(x_i) = f(x'_i)$ for some $i \in [d]$, which will complete the proof Theorem 8.

In the rest of this proof, we will show that if $W_t \otimes x_i \neq W_t \otimes x'_i$, then (a) or (b) occurs. Without loss of generality, we assume

$$W_1 \otimes x_i \neq W_1 \otimes x'_i \quad (6)$$

We first claim the following statement.

Claim 16. *If $|W_{1j}| > 2^{\epsilon_{\max} - e^* + 1}$ for some $j \in [d]$, then (a) or (b) occurs.*

Proof. Suppose that $|W_{1j}| > 2^{\epsilon_{\max} - e^* + 1}$ for some $j \in [d]$. Then, since

$$|W_{1j}| \otimes (2^{e^*})^- \geq (2^{\epsilon_{\max} - e^* + 1})^+ \otimes (2^{e^*})^- = \infty,$$

we have $W_{1j} \otimes x_{kj} = W_{1j} \otimes x'_{kj} \in \{\pm\infty\}$ for all $k \neq j$. This implies that for each $k \neq j$,

- $W_1 \otimes x_k = W_1 \otimes x'_k \in \{\pm\infty\}$ or
- $W_1 \otimes x_k = \text{NaN}$ or $W_1 \otimes x'_k = \text{NaN}$.

Hence, if $j = i$, then (a) or (b) occurs. If $j \neq i$, then we cannot have $W_1 \otimes x_i = W_1 \otimes x'_i$ by the assumption Eq. (6), which implies $W_1 \otimes x_i = \text{NaN}$ or $W_1 \otimes x'_i = \text{NaN}$, i.e., (b) occurs. Thus, the claim is valid. $\square$

Given Claim 16, we now assume $|W_{1j}| \leq 2^{\epsilon_{\max} - e^* + 1}$ for all $j \in [d]$. For each $l \in [d]$, we define

$$h_l(z_1, \dots, z_d) = \bigoplus_{j=1}^l (W_{1j} \otimes z_j).$$

Since $W_1 \otimes x_i \neq W_1 \otimes x'_i$ by the assumption Eq. (6) (i.e., $h_d(x_i) \neq h_d(x'_i)$), we must have $h_i(x_i) \neq h_i(x'_i)$. We next claim the following statement.

Claim 17. $W_1 \otimes x_k = W_1 \otimes x'_k$ for all $k > i$.

Proof. By Claim 16, $|W_{1i}| \leq 2^{\epsilon_{\max} - e^* + 1}$. Then, since

$$|W_{1i} \otimes x_{ii}|, |W_{1i} \otimes x'_{ii}| \leq |W_{1i}| \otimes \omega \leq 2^{\epsilon_{\min} - p + \epsilon_{\max} - e^* + 1},$$

to have $h_i(x_i) \neq h_i(x'_i)$, we must have $|h_{i-1}(x_i)| \leq 2^{\epsilon_{\min} + \epsilon_{\max} - e^* + 3}$. This also implies

$$|h_i(x_i)|, |h_i(x'_i)| \leq 2^{\epsilon_{\min} + \epsilon_{\max} - e^* + 3}.$$

Furthermore, since $\omega \otimes (2^{e^*})^- = (2^{e^* + \epsilon_{\min} - p})^-$ and $(e^* + \epsilon_{\min} - p) - p - 2 > \epsilon_{\min} + \epsilon_{\max} - e^* + 3$ by the definition of e^* , we have

$$h_i(x_i) \oplus (W_{1k} \otimes x_{ik}) = W_{1k} \otimes x_{ik} = h_i(x'_i) \oplus (W_{1k} \otimes x'_{ik})$$

for all $W_{1k} \in \mathbb{F} \setminus \{0\}$ and $k > i$. Hence, to have $W_1 \otimes x_i \neq W_1 \otimes x'_i$, $W_{1k} = 0$ for all $k > i$, i.e., $W_1 \otimes x_k = W_1 \otimes x'_k$ for all $k > i$. Thus, the claim is valid. $\square$

Lastly, we cover the remaining case in the following claim.

Claim 18. $W_1 \otimes x_k = W_1 \otimes x'_k$ for all $k < i$.

Proof. Suppose that $W_1 \otimes x_k \neq W_1 \otimes x'_k$ for some $k < i$. Then, as we argued in the proof of Claim 17, we must have $W_{1l} = 0$ for all $l > k$. However, this is not possible since $i > k$ but $W_{1i} \neq 0$ by the assumption Eq. (6). Hence, the claim is valid. $\square$

5. Conclusion

In this work, we present upper and lower bounds on the minimum depth and width of floating-point networks for representing floating-point functions. We first show that the minimum depth is exactly three, which is strictly larger than that under exact arithmetic. However, two layers are sufficient if we consider a small domain or codomain. We also show that the minimum width lies between $2d$ and $2d + 4$ in general, while it is approximately d under exact arithmetic. As in the minimum depth case, we prove that the minimum width decreases as the domain gets smaller, even beyond d . Our results show a clear discrepancy between neural networks under floating-point arithmetic and exact arithmetic. We believe our results help to better understand real implementations of neural networks.

Acknowledgement

This work was supported by the National Research Foundation of Korea (NRF) grant (No. RS-2024-00406127, RS-2024-00415812, RS-2025-00515264, RS-2025-24534003, RS-2026-25472534, RS-2026-25615334), Institute of Information & communications Technology Planning & Evaluation (IITP) grant (No. RS-2019-II190079, 2019-0-01842) funded by the Korea government (MSIT), and Global University Project grant funded by the GIST in 2025.

Impact Statement

This paper analyzes the representation power of neural networks under floating-point arithmetic. We could not find notable potential societal consequences of our work.

References

- Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., Corrado, G. S., Davis, A., Dean, J., Devin, M., et al. Tensorflow: Large-scale machine learning on heterogeneous distributed systems. *arXiv preprint arXiv:1603.04467*, 2016.
- Baum, E. B. On the capabilities of multilayer perceptrons. *Journal of complexity*, 4(3):193–215, 1988.
- Bubeck, S., Eldan, R., Lee, Y. T., and Mikulincer, D. Network size and size of the weights in memorization with two-layers neural networks. In *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2020.
- Cai, Y. Achieve the minimum width of neural networks for universal approximation. In *International Conference on Learning Representations (ICLR)*, 2023.
- Cybenko, G. Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals, and Systems*, 2(4):303–314, 1989.
- Google. Improve your model’s performance with bfloat16. <https://cloud.google.com/tpu/docs/bfloat16>.
- Hanin, B. and Sellke, M. Approximating continuous functions by ReLU nets of minimal width. *arXiv preprint arXiv:1710.11278*, 2017.
- Hornik, K., Stinchcombe, M., and White, H. Multilayer feedforward networks are universal approximators. *Neural Networks*, 2(5):359–366, 1989.
- Huang, G.-B. Learning capability and storage capacity of two-hidden-layer feedforward networks. *IEEE Transactions on Neural Networks*, 14(2):274–281, 2003.
- Huang, G.-B. and Babri, H. A. Upper bounds on the number of hidden neurons in feedforward networks with arbitrary bounded nonlinear activation functions. *IEEE Transactions on Neural Networks*, 9(1):224–229, 1998.
- Hwang, G., Lee, W., Park, Y., Park, S., and Saad, F. Floating-Point Neural Networks Are Provably Robust Universal Approximators. In *International Conference on Computer Aided Verification (CAV)*, 2025a.
- Hwang, G., Park, Y., Lee, W., and Park, S. Floating-point neural networks can represent almost all floating-point functions. In *International Conference on Machine Learning (ICML)*, 2025b.
- IEEE. IEEE Standard for Floating-Point Arithmetic. *IEEE Std 754-2019 (Revision of IEEE 754-2008)*, pp. 1–84, 2019. doi: 10.1109/IEEESTD.2019.8766229.
- Johnson, J. Deep, skinny neural networks are not universal approximators. In *International Conference on Learning Representations (ICLR)*, 2019.
- Kidger, P. and Lyons, T. Universal approximation with deep narrow networks. In *Conference on Learning Theory (COLT)*, 2020.
- Kim, N., Min, C., and Park, S. Minimum width for universal approximation using ReLU networks on compact domain. In *International Conference on Learning Representations (ICLR)*, 2024.
- Leshno, M., Lin, V. Y., Pinkus, A., and Schocken, S. Multilayer feedforward networks with a nonpolynomial activation function can approximate any function. *Neural Networks*, 6(6):861–867, 1993.
- Lu, Z., Pu, H., Wang, F., Hu, Z., and Wang, L. The expressive power of neural networks: A view from the width. In *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2017.
- Micikevicius, P., Stosic, D., Burgess, N., Cornea, M., Dubey, P., Grisenthwaite, R., Ha, S., Heinecke, A., Judd, P., Kamalu, J., et al. FP8 Formats for Deep Learning. *arXiv preprint arXiv:2209.05433*, 2022.
- Park, S., Yun, C., Lee, J., and Shin, J. Minimum width for universal approximation. In *International Conference on Learning Representations (ICLR)*, 2021.
- Park, Y., Hwang, G., Lee, W., and Park, S. Expressive power of ReLU and step networks under floating-point operations. *Neural Networks*, 175:106297, 2024.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., et al. Pytorch: An imperative style, high-performance

deep learning library. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2019.

Pinkus, A. Approximation theory of the mlp model in neural networks. *Acta numerica*, 8:143–195, 1999.

Shin, J., Kim, N., Hwang, G., and Park, S. Minimum Width for Universal Approximation using Squashable Activation Functions. In *International Conference on Machine Learning (ICML)*, 2025.

A. Values of p and q in Practical Floating-Point Formats

 Table 2. Values of p and q in practical floating-point formats.

Floating-point format	(p, q)
64-bit double-precision (IEEE, 2019)	(52, 11)
32-bit single-precision (IEEE, 2019)	(23, 8)
16-bit half-precision (IEEE, 2019)	(10, 5)
bfloat16 (Google)	(7, 8)
8-bit E5M2 (Micikevicius et al., 2022)	(2, 5)
8-bit E4M3 (Micikevicius et al., 2022)	(3, 4)

B. Technical lemmas

Lemma 19. *Let $z \in (0, 1]_{\mathbb{F}}$, $e \in \mathbb{Z}$ such that $z - z^- = 2^e$, and $k \in [\epsilon_{\min} - p, \epsilon_{\max} + e - 1]_{\mathbb{Z}}$. Then, there exist $u, w, b \in \mathbb{F}$ such that*

$$g(x) = u \otimes \sigma(w \otimes x \oplus b) = \begin{cases} 2^k & \text{if } x = z, \\ 0 & \text{if } x \in [-1, z]_{\mathbb{F}}, \end{cases}$$

and $g(x) \in [0, 2^{\epsilon_{\max} + e_x - 1}]_{\mathbb{F}} \subset \mathbb{F}$ for all $x \in (z, 1]_{\mathbb{F}}$ where e_x denotes the integer satisfying $2^{e_x - 1} < x \leq 2^{e_x}$.

Proof. We choose $b = -z^-$, $w = 1$, and $u = 2^{k-e}$. Here, $u \in \mathbb{F}$ since $e \leq -p - 1$, i.e., $k - e \in [\epsilon_{\min} + 1, \epsilon_{\max} - 1]_{\mathbb{Z}}$. Then, one can observe that $g(x) = 0$ if $x \in [-1, z]_{\mathbb{F}}$ and $g(z) = 2^k$. Furthermore, from our choices of b, w, u , it holds that for each $x \in (z, 1]_{\mathbb{F}}$,

$$0 \leq g(x) \leq u \otimes \sigma(w \otimes 2^{e_x}) = 2^{k-e+e_x}$$

Since $e_x \leq 0$ for all $x \in (0, 1]_{\mathbb{F}}$ and $k - e \leq \epsilon_{\max} - 1$, we also have $2^{k-e+e_x} \leq 2^{\epsilon_{\max} - 1 + e_x} < \infty$. This completes the proof. $\square$

Lemma 20. *Let $z \in (0, 1]_{\mathbb{F}}$, $e, e_z \in \mathbb{Z}$ such that $z - z^- = 2^e$ and $2^{e_z - 1} < z \leq 2^{e_z}$, and $\gamma = \max\{1, 2^{\epsilon_{\max} + e_z - 1}\}$. Then, for any $y \in [0, \gamma]_{\mathbb{F}}$, there exist $n \in \mathbb{N}$ and $e_1, \dots, e_n \in [\epsilon_{\min} - p, \epsilon_{\max} + e - 1]_{\mathbb{Z}}$ such that $\bigoplus_{i=1}^n 2^{e_i} = y$ and $y \oplus \bigoplus_{i=1}^n (-2^{e_i}) = 0$.*

Proof. We first consider the case that $\gamma = 1$. Since $e \geq \epsilon_{\min} - p$ and $\epsilon_{\max} + e - 1 \geq \epsilon_{\max} + \epsilon_{\min} - p - 1 = -p$, for any $y \in [0, 1]_{\mathbb{F}}$, there exist $n \in [2^{p+1}]$ and $e^* \in [\epsilon_{\min} - p, -p]_{\mathbb{Z}} \subset [\epsilon_{\min} - p, \epsilon_{\max} + e - 1]_{\mathbb{Z}}$ such that $\bigoplus_{i=1}^n 2^{e^*} = y$ and $y \oplus \bigoplus_{i=1}^n (-2^{e^*}) = 0$. Choosing $e_1, \dots, e_n = e^*$ completes the proof for the case $\gamma = 1$.

We now consider the case that $\gamma = 2^{\epsilon_{\max} + e_z - 1}$. If $e_z \geq \epsilon_{\min} + 2$, then $e = e_z - p - 1$. Furthermore, for any $y \in [0, 2^{\epsilon_{\max} + e_z - 1}]_{\mathbb{F}}$, there exist $n \in [2^{p+1}]$ and $e^* \in [\epsilon_{\min} - p, \epsilon_{\max} + e_z - p - 2]_{\mathbb{Z}} \subset [\epsilon_{\min} - p, \epsilon_{\max} + e - 1]_{\mathbb{Z}}$ such that $\bigoplus_{i=1}^n 2^{e^*} = y$ and $y \oplus \bigoplus_{i=1}^n (-2^{e^*}) = 0$. Choosing $e_1, \dots, e_n = e^*$ suffices for the proof. If $e_z \leq \epsilon_{\min} + 1$, then $e = \epsilon_{\min} - p$ and $\gamma \leq 2$, i.e., $y \in [0, 2]_{\mathbb{F}}$. Hence, one can find $n \in [2^{p+1}]$ and $e_1, \dots, e_n \in [\epsilon_{\min} - p, \epsilon_{\max} + e - 1]_{\mathbb{Z}} = [\epsilon_{\min} - p, -p]_{\mathbb{Z}}$ such that $\bigoplus_{i=1}^n 2^{e_i} = y$ and $y \oplus \bigoplus_{i=1}^n (-2^{e_i}) = 0$. This completes the proof. $\square$

Lemma 21. *Let $z \in (0, 1]_{\mathbb{F}}$. Then, for any $g : [-1, 1]_{\mathbb{F}} \rightarrow \mathbb{F}$ satisfying $g(x) \in [-\gamma_x, \gamma_x]_{\mathbb{F}}$ for all $x \in [z, 1]_{\mathbb{F}}$ where $\gamma_x = \max\{1, 2^{\epsilon_{\max} + e_x - 1}\}$ and e_x denotes the integer satisfying $2^{e_x - 1} < x \leq 2^{e_x}$, there exist $n \in \mathbb{N}$ and $u_i, w_i, b_i \in \mathbb{F}$ for all $i \in [n]$ such that*

$$h(x) = g(x) \oplus \bigoplus_{i=1}^n (u_i \otimes \sigma(w_i \otimes x \oplus b_i)) = \begin{cases} g(x) & \text{if } x \in [-1, z]_{\mathbb{F}}, \\ 0 & \text{if } x \in [z, 1]_{\mathbb{F}}, \end{cases}$$

for all $x \in [-1, 1]_{\mathbb{F}}$.

Proof. Let $\omega = z_\gamma < z_{m-1} < \dots < z_1 = 1$ be all floats between ω and 1. We inductively prove that the statement of Lemma 21 is true for $z = z_m$ for some $m \in [\gamma]$. Consider the base case $m = 1$ where $z = 1$. Consider $g : [-1, 1]_{\mathbb{F}} \rightarrow \mathbb{F}$ satisfying $g(1) \in [-\gamma_1, \gamma_1]_{\mathbb{F}}$ where $\gamma_1 = 2^{\epsilon_{\max}-1}$. Without loss of generality, we assume $g(1) > 0$. Choose $e = -p - 1$ and $e_z = 0$ so that $2^e = z - z^-$ and $2^{e_z-1} < z \leq 2^{e_z}$. Then, by Lemma 20, there exist $n \in \mathbb{N}$ and $e_1, \dots, e_n \in [\epsilon_{\min} - p, \epsilon_{\max} + e - 1]_{\mathbb{Z}}$ such that $g(1) \oplus \bigoplus_{i=1}^n (-2^{e_i}) = 0$. Furthermore, by Lemma 19, there exist $u_1, w_1, b_1, \dots, u_n, w_n, b_n \in \mathbb{F}$ such that for each $i \in [n]$,

$$u_i \otimes \sigma(w_i \otimes x \oplus b_i) = \begin{cases} 2^{e_i} & \text{if } x = 1, \\ 0 & \text{if } x \in [-1, 1)_{\mathbb{F}}. \end{cases}$$

We choose

$$h(x) = g(x) \oplus \bigoplus_{i=1}^n ((-u_i) \otimes \sigma(w_i \otimes x \oplus b_i)) = \begin{cases} g(x) & \text{if } x \in [-1, 1)_{\mathbb{F}}, \\ 0 & \text{if } x = 1, \end{cases}$$

which completes the proof for the base case $z = 1$ (i.e., $m = 1$).

We now consider general $m > 1$ (i.e., $z = z_m$). Consider $g : [-1, 1]_{\mathbb{F}} \rightarrow \mathbb{F}$ satisfying $g(x) \in [-\gamma_x, \gamma_x]_{\mathbb{F}}$ for all $x \in [z_m, 1]_{\mathbb{F}}$. Without loss of generality, we assume $g(z_m) > 0$. Choose $e, e_z \in \mathbb{Z}$ such that $2^e = z_m - z_m^-$ and $2^{e_z-1} < z_m \leq 2^{e_z}$. Then, by Lemma 20, there exist $l \in \mathbb{N}$ and $e_1, \dots, e_l \in [\epsilon_{\min} - p, \epsilon_{\max} + e - 1]_{\mathbb{Z}}$ such that $g(z_m) \oplus \bigoplus_{i=1}^l (-2^{e_i}) = 0$. Furthermore, by Lemma 19, there exist $u_1, w_1, b_1, \dots, u_l, w_l, b_l \in \mathbb{F}$ such that for each $i \in [l]$,

$$\begin{aligned} u_i \otimes \sigma(w_i \otimes x \oplus b_i) &= \begin{cases} 2^{e_i} & \text{if } x = z_m, \\ 0 & \text{if } x \in [-1, z_m)_{\mathbb{F}}, \end{cases} \\ u_i \otimes \sigma(w_i \otimes x \oplus b_i) &\in [0, 2^{\epsilon_{\max}-1+e_x}]_{\mathbb{F}} \subset [0, \gamma_x]_{\mathbb{F}} \text{ if } x \in (z, 1]_{\mathbb{F}}. \end{aligned} \quad (7)$$

Let $g_0 = g$. We inductively define h_i and g_j for each $i \in [l+1]$ and $j \in [l]$ as follows:

$$\begin{aligned} h_i(x) &= g_{i-1}(x) \oplus \bigoplus_{k=1}^{r_i} (u'_{i,k} \otimes \sigma(w'_{i,k} \otimes x \oplus b'_{i,k})), \\ g_j(x) &= h_j(x) \oplus ((-u_j) \otimes \sigma(w_j \otimes x \oplus b_j)), \end{aligned}$$

where $r_i \in \mathbb{N}$ and $u'_{i,k}, w'_{i,k}, b'_{i,k} \in \mathbb{F}$ are chosen by using the induction hypothesis, so that

$$h_i(x) = \begin{cases} g_{i-1}(x) & \text{if } x \in [-1, z_m]_{\mathbb{F}}, \\ 0 & \text{if } x \in (z_m, 1]_{\mathbb{F}}. \end{cases}$$

We note that we can use the induction hypothesis here due to Eq. (7). Then, by the definitions of h_i and g_j , one can observe that

$$h_i(x) = \begin{cases} g(x) & \text{if } x \in [-1, z_m)_{\mathbb{F}}, \\ g(z_m) \oplus \bigoplus_{k=1}^{i-1} (-2^{e_k}) & \text{if } x = z_m, \\ 0 & \text{if } x \in (z_m, 1]_{\mathbb{F}}. \end{cases}$$

Hence, $h = h_{l+1}$ satisfies the desired property. This completes the proof. $\square$

Lemma 22. Let $g : [-1, 1]_{\mathbb{F}} \rightarrow \mathbb{F}$ be a function satisfying $g(x) \in [-\gamma_x, \gamma_x]_{\mathbb{F}}$ for all $x \in (0, 1]_{\mathbb{F}}$ where $\gamma_x = \max\{1, 2^{\epsilon_{\max}+e_x-1}\}$ and e_x denotes the integer satisfying $2^{e_x-1} < x \leq 2^{e_x}$. Let $f^* : (0, 1]_{\mathbb{F}} \rightarrow [-1, 1]_{\mathbb{F}}$. Then, there exist $n \in \mathbb{N}$ and $u_i, w_i, b_i \in \mathbb{F}$ for all $i \in [n]$ such that for any $x \in [-1, 1]_{\mathbb{F}}$,

$$h(x) = g(x) \oplus \bigoplus_{i=1}^n (u_i \otimes \sigma(w_i \otimes x \oplus b_i)) = \begin{cases} g(x) & \text{if } x \in [-1, 0]_{\mathbb{F}}, \\ f^*(x) & \text{if } x \in (0, 1]_{\mathbb{F}}. \end{cases}$$

Proof. Let $0 = z_1 < z_2 < \dots < z_\gamma = 1$ be all floats in $[0, 1]_{\mathbb{F}}$. We inductively show that for each $m \in [\gamma]$, there exist $n_i \in \mathbb{N}$, $u_{i,j}, w_{i,j}, b_{i,j} \in \mathbb{F}$ for all $j \in [n_i]$, and

$$h_i(x) = g(x) \oplus \bigoplus_{j=1}^{n_i} u_{i,j} \otimes \sigma(w_{i,j} \otimes x \oplus b_{i,j})$$

such that

$$h_i(x) = \begin{cases} g(x) & \text{if } x \in [-1, 0]_{\mathbb{F}}, \\ f^*(x) & \text{if } x \in (0, z_i]_{\mathbb{F}}, \\ 0 & \text{if } x \in (z_i, 1]_{\mathbb{F}}. \end{cases} \quad (8)$$

Then, $h = h_m$ satisfies the desired properties.

Consider the base case: $m = 1$. By Lemma 21, there exist $r \in \mathbb{N}$ and $u'_i, w'_i, b'_i \in \mathbb{F}$ such that

$$h_1(x) = g(x) \oplus \bigoplus_{i=1}^r (u'_i \otimes \sigma(w'_i \otimes x \oplus b'_i)) = \begin{cases} g(x) & \text{if } x \in [-1, 0]_{\mathbb{F}}, \\ 0 & \text{if } x \in (0, 1]_{\mathbb{F}}. \end{cases}$$

This proves the base case.

Now, consider the general case: $m > 1$. Let $e, e_{z_m} \in \mathbb{Z}$ such that $2^e = z_m - z_m^-$ and $2^{e_{z_m}-1} < z_m \leq 2^{e_{z_m}}$. Choose h_{m-1} satisfying Eq. (8); its existence is guaranteed by the induction hypothesis. By Lemma 20, there exist $l \in \mathbb{N}$ and $e_1, \dots, e_l \in [\mathfrak{e}_{\min} - p, \mathfrak{e}_{\max} + e - 1]$ such that $|f^*(z_m)| = \bigoplus_{i=1}^l 2^{e_i}$ since $f^*(z_m) \in [-1, 1]_{\mathbb{F}}$. Furthermore, by Lemma 19, there exist $u_i, w_i, b_i \in \mathbb{F}$ for all $i \in [l]$ such that

$$\begin{aligned} u_i \otimes \sigma(w_i \otimes x \oplus b_i) &= \begin{cases} 2^{e_i} & \text{if } x = z_m, \\ 0 & \text{if } x \in [-1, z_m)_{\mathbb{F}}, \end{cases} \\ u_i \otimes \sigma(w_i \otimes x \oplus b_i) &\in [0, \gamma_x] \quad \text{if } x \in (z_m, 1]_{\mathbb{F}}. \end{aligned} \quad (9)$$

Let $g_0 = h_{m-1}$. To construct h_m , we inductively define f_i, g_i for all $i \in [l]$ as follows: for $s \in \{-1, 1\}$ such that $s \times f^*(z_m) \geq 0$,

$$\begin{aligned} f_i(x) &= g_{i-1}(x) \oplus ((su_i) \otimes \sigma(w_i \otimes x \oplus b_i)), \\ g_i(x) &= f_i(x) \oplus \bigoplus_{k=1}^{r_i} (u'_{i,k} \otimes \sigma(w'_{i,k} \otimes x \oplus b'_{i,k})), \end{aligned}$$

where $r_i \in \mathbb{N}$ and $u'_{i,k}, w'_{i,k}, b'_{i,k} \in \mathbb{F}$ are chosen by using Lemma 21, so that

$$g_i(x) = \begin{cases} f_i(x) & \text{if } x \in [-1, z_m]_{\mathbb{F}}, \\ 0 & \text{if } x \in (z_m, 1]_{\mathbb{F}}, \end{cases}$$

for all $i \in [l]$. We note that we can apply Lemma 21 due to Eqs. (8) and (9). From the definitions of f_i, g_i , and by Eq. (9), one can observe that for each $i \in [l] \cup \{0\}$,

$$g_i(x) = \begin{cases} g(x) & \text{if } x \in [-1, 0]_{\mathbb{F}}, \\ f^*(x) & \text{if } x \in (0, z_m)_{\mathbb{F}}, \\ \bigoplus_{j=1}^i ((su_j) \otimes \sigma(w_j \otimes x \oplus b_j)) & \text{if } x = z_m, \\ 0 & \text{if } x \in (z_m, 1]_{\mathbb{F}}. \end{cases}$$

Hence, choosing $h_m = g_l$ completes the proof. $\square$

Lemma 23. Let $z \in [0, \Omega)_{\mathbb{F}}$ and $v \in \{2^e : e \in [\mathfrak{e}_{\min} - p, \mathfrak{e}_{\max} - p - 1]_{\mathbb{Z}}\} \cup \{(2^{\mathfrak{e}_{\max}-p-1})^+\}$. Then, there exist $u, w, b \in \mathbb{F}$ such that

$$\begin{aligned} u \otimes \sigma(w \otimes x \oplus b) &= \begin{cases} 0 & \text{if } x \in (z, \Omega]_{\mathbb{F}}, \\ v & \text{if } x = z, \end{cases} \\ u \otimes \sigma(w \otimes x \oplus b) &\in [0, (2^{\mathfrak{e}_{\max}})^+]_{\mathbb{F}} \quad \text{if } x \in [0, z)_{\mathbb{F}}. \end{aligned}$$

Proof. Let $e_z \in \mathbb{Z}$ such that $2^{e_z-1} \leq z < 2^{e_z}$. We first consider $z \geq 2^{\epsilon_{\min}}$, i.e., $e_z > \epsilon_{\min}$. In this case, we choose $w = -2^{-e_z+1}$ and $b = 2^{-e_z+1} \times z^+$; we note that $-2^{-e_z+1}, 2^{-e_z+1} \times z^+ \in \mathbb{F}$ since $e_z > \epsilon_{\min}$ and $p \geq 2$. Then, we have

$$\begin{aligned} \sigma(w \otimes x \oplus b) &= \begin{cases} 0 & \text{if } x \in (z, \Omega]_{\mathbb{F}}, \\ 2^{-p} & \text{if } x = z, \end{cases} \\ \sigma(w \otimes x \oplus b) &\in [0, 2]_{\mathbb{F}} \text{ if } x \in [0, z)_{\mathbb{F}}. \end{aligned}$$

If $v = 2^e$ for some $e \in [\epsilon_{\min} - p, \epsilon_{\max} - p - 1]_{\mathbb{Z}}$, then choose $u = 2^{e+p} \in \mathbb{F}$. If $v = (2^{\epsilon_{\max}-p-1})^+$, then choose $u = (2^{\epsilon_{\max}-1})^+ \in \mathbb{F}$. Under these choices of u, w, b , we have

$$\begin{aligned} \sigma(w \otimes x \oplus b) &= \begin{cases} 0 & \text{if } x \in (z, \Omega]_{\mathbb{F}}, \\ v & \text{if } x = z, \end{cases} \\ \sigma(w \otimes x \oplus b) &\in [0, (2^{\epsilon_{\max}})^+]_{\mathbb{F}} \text{ if } x \in [0, z)_{\mathbb{F}}. \end{aligned}$$

If $z < 2^{\epsilon_{\min}}$, then we choose $w = -2^{-\epsilon_{\min}} = -2^{\epsilon_{\max}-1}$ and $b = 2^{-\epsilon_{\min}} \times z^+ \leq 1$. Then, it holds that

$$\begin{aligned} \sigma(w \otimes x \oplus b) &= \begin{cases} 0 & \text{if } x \in (z, \Omega]_{\mathbb{F}}, \\ 2^{-p} & \text{if } x = z, \end{cases} \\ \sigma(w \otimes x \oplus b) &\in [0, b]_{\mathbb{F}} \subset [0, 1]_{\mathbb{F}} \text{ if } x \in [0, z)_{\mathbb{F}}. \end{aligned}$$

If $v = 2^e$ for some $e \in [\epsilon_{\min} - p, \epsilon_{\max} - p - 1]_{\mathbb{Z}}$, then choose $u = 2^{e+p} \in \mathbb{F}$. If $v = (2^{\epsilon_{\max}-p-1})^+$, then choose $u = (2^{\epsilon_{\max}-1})^+ \in \mathbb{F}$. Under these choices of u, w, b , we have

$$\begin{aligned} \sigma(w \otimes x \oplus b) &= \begin{cases} 0 & \text{if } x \in (z, \Omega]_{\mathbb{F}}, \\ v & \text{if } x = z, \end{cases} \\ \sigma(w \otimes x \oplus b) &\in [0, (2^{\epsilon_{\max}-1})^+]_{\mathbb{F}} \text{ if } x \in [0, z)_{\mathbb{F}}. \end{aligned}$$

This completes the proof. $\square$

Lemma 24. For any $z_1 \in [0, \infty]_{\mathbb{F}}$ and $z_2 \in [0, \Omega]_{\mathbb{F}}$, there exist $n_1, n_2 \in \mathbb{N}$ and $v_1, \dots, v_{n_1}, u_1, \dots, u_{n_2} \in \{2^e : e \in [\epsilon_{\min} - p, \epsilon_{\max} - p - 1]_{\mathbb{Z}}\} \cup \{(2^{\epsilon_{\max}-p-1})^+\}$ such that $\bigoplus_{i=1}^{n_1} v_i = z_1$ and $z_2 \oplus \bigoplus_{i=1}^{n_2} (-u_i) = 0$.

Proof. We first consider $z = z_1 = z_2 \in [0, \Omega]_{\mathbb{F}}$. For $z = 0$, choose $n_1 = n_2 = 0$. For $z \in (0, 2^{\epsilon_{\max}}]_{\mathbb{F}}$, there exist $n \in [2^{p+1}]$ and $e^* \in [\epsilon_{\min} - p, \epsilon_{\max} - p - 1]_{\mathbb{Z}}$ such that $z = \bigoplus_{i=1}^n 2^{e^*}$ and $z \oplus \bigoplus_{i=1}^n (-2^{e^*}) = 0$. Choose $n_1 = n_2 = n$ and $v_1 = \dots, v_{n_1} = u_1 = \dots, u_{n_2} = 2^{e^*}$. Let $(2^{\epsilon_{\max}})^+ = z_1 < z_2 < \dots < z_{2^p-1} = \Omega$ be all floats between $(2^{\epsilon_{\max}})^+$ and Ω . For $z = z_j$ for some $j \in [2^p - 1]$, let $n_1 = n_2 = 2^{p+1} + j$, $v_1 = \dots = v_{2^p+1} = u_{j+1} = \dots = u_{n_2} = 2^{\epsilon_{\max}-p-1}$, and $v_{2^p+1+1} = \dots = v_{n_1} = u_1 = \dots = u_j = (2^{\epsilon_{\max}-p-1})^+$. Then, $z_j = \bigoplus_{i=1}^{n_1} v_i$ and $z \oplus \bigoplus_{i=1}^{n_2} (-u_i) = 0$.

Now, consider $z_1 = \infty$. In this case, choose $n_1 = 3 \times 2^p$, $v_1 = \dots = v_{2 \times 2^p} = 2^{\epsilon_{\max}-p-1}$, and $v_{2 \times 2^p+1} = \dots = v_{3 \times 2^p} = (2^{\epsilon_{\max}-p-1})^+$. Then, $\bigoplus_{i=1}^{n_1} v_i = \infty = z_1$. This completes the proof. $\square$

Lemma 25. Let $z \in [0, \Omega]_{\mathbb{F}}$. Then, for any $g : [0, \Omega] \rightarrow \mathbb{F} \cup \{\pm\infty\}$ such that $g([0, z]_{\mathbb{F}}) \subset \mathbb{F}$, there exist $n \in \mathbb{N}$ and $u_i, w_i, b_i \in \mathbb{F}$ for all $i \in [n]$ such that

$$h(x) = g(x) \oplus \bigoplus_{i=1}^n (u_i \otimes \sigma(w_i \otimes x \oplus b_i)) = \begin{cases} 0 & \text{if } x \in [0, z]_{\mathbb{F}}, \\ g(x) & \text{if } x \in (z, \Omega]_{\mathbb{F}}, \end{cases}$$

for all $x \in [0, \Omega]_{\mathbb{F}}$.

Proof. Let $0 = z_1 < z_2 < \dots < z_{\gamma} = \Omega^-$ be all floats between 0 and Ω^- and

$$\mathcal{V} = \{2^e : e \in [\epsilon_{\min} - p, \epsilon_{\max} - p - 1]_{\mathbb{Z}}\} \cup \{(2^{\epsilon_{\max}-p-1})^+\}.$$

Let $z = z_m$. Then, by Lemma 24, there exists $l \in \mathbb{N}$ and $v_1, \dots, v_l \in \mathcal{V}$ such that $g(z_m) \oplus \bigoplus_{i=1}^l (-v_i) = 0$. Furthermore, by Lemma 23, there exist $u_i, w_i, b_i \in \mathbb{F}$ such that

$$\begin{aligned} u_i \otimes \sigma(w_i \otimes x \oplus b) &= \begin{cases} v_i & \text{if } x = z_m, \\ 0 & \text{if } x \in (z_m, \Omega]_{\mathbb{F}}, \end{cases} \\ u_i \otimes \sigma(w_i \otimes x \oplus b) &\in \mathbb{F} & \text{if } x \in [0, z_m)_{\mathbb{F}}, \end{aligned} \quad (10)$$

for all $i \in [l]$.

We prove Lemma 25 using mathematical induction on m . Without loss of generality, we assume $g(z_m) > 0$. For the base case $m = 1$ (i.e., $z = 0$), we have

$$h(x) = g(x) \oplus \bigoplus_{i=1}^l ((-u_i) \otimes \sigma(w_i \otimes x \oplus b)) = \begin{cases} 0 & \text{if } x = 0, \\ g(x) & \text{if } x \in (0, \Omega]_{\mathbb{F}}. \end{cases}$$

Now, consider the general case: $m > 1$. Let $g_0 = g$. To prove the general case, we inductively construct h_i and g_j for each $i \in [l+1]$ and $j \in [l]$ as follows:

$$\begin{aligned} h_i(x) &= g_{i-1}(x) \oplus \bigoplus_{k=1}^{r_i} (u'_{i,k} \otimes \sigma(w'_{i,k} \otimes x \oplus b'_{i,k})), \\ g_j(x) &= h_j(x) \oplus ((-u_j) \otimes \sigma(w_j \otimes x \oplus b_j)), \end{aligned}$$

where $r_i \in \mathbb{N}$ and $u'_{i,k}, w'_{i,k}, b'_{i,k} \in \mathbb{F}$ are chosen by using the induction hypothesis and Eq. (10) so that

$$h_i(x) = \begin{cases} 0 & \text{if } x \in [0, z_{m-1}]_{\mathbb{F}}, \\ g_{i-1}(x) & \text{if } x \in (z_{m-1}, \Omega]_{\mathbb{F}}. \end{cases}$$

Then, by the definitions of h_i and g_j , we have

$$h_i(x) = \begin{cases} 0 & \text{if } x \in [0, z_m)_{\mathbb{F}}, \\ g(z_m) \oplus \bigoplus_{k=1}^{i-1} (-v_k) & \text{if } x = z_m, \\ g(x) & \text{if } x \in (z_m, \Omega]_{\mathbb{F}}. \end{cases}$$

Hence, choosing $h = h_{l+1}$ completes the proof. $\square$

Lemma 26. Let $d, d' \in \mathbb{N}$, $W \in \mathbb{F}^{d' \times d}$, and $b \in \mathbb{F}^{d'}$. Let $\alpha \in \{\pm\infty, \text{NaN}\}$ and $\mathcal{X} = \{x = (x_1, \dots, x_d) \in \overline{\mathbb{F}}^d : x_1 = \alpha, (W \otimes x \oplus b)_j \neq \text{NaN} \forall j \in [d']\}$. Then, $W \otimes x \oplus b$ is identical for all $x \in \mathcal{X}$.

Proof. Since the proof is trivial if $\alpha = \text{NaN}$, without loss of generality, suppose $\alpha = \infty$. Since $(W \otimes x \oplus b)_j \neq \text{NaN}$ for all $x \in \mathcal{X}$ and $j \in [d']$, all $W_{11}, \dots, W_{d'1}$ should be non-zero. Let $s_j = 1$ if $W_{j1} > 0$ and $s_j = -1$ if $W_{j1} < 0$ for all $j \in [d']$. Then, one can observe that $W \otimes x \oplus b$ should be $(s_1 \otimes \infty, \dots, s_{d'} \otimes \infty)$. This completes the proof. $\square$

C. Additional proofs

C.1. Proof of Theorem 3

Let $h_0(x) = f^*(0) \otimes \text{ReLU}(0 \otimes x \oplus 1)$. Then, $h_0(x) = f^*(0) \in [-1, 1]_{\mathbb{F}}$ for all $x \in [-1, 1]_{\mathbb{F}}$. By Lemma 22, there exist $n_1 \in \mathbb{N}$ and $u_i, w_i, b_i \in \mathbb{F}$ for all $i \in [n_1]$ such that

$$h_1(x) = h_0(x) \oplus \bigoplus_{i=1}^{n_1} (u_i \otimes \sigma(w_i \otimes x \oplus b_i)) = \begin{cases} h_0(x) & \text{if } x \in [-1, 0]_{\mathbb{F}}, \\ f^*(x) & \text{if } x \in (0, 1]_{\mathbb{F}}, \end{cases}$$

i.e., $h_1(x) = f^*(x)$ for $x \in [0, 1]_{\mathbb{F}}$ and $h_1(x) \in [-1, 1]_{\mathbb{F}}$ for $x \in [-1, 0]_{\mathbb{F}}$. Again, by Lemma 22 and symmetry, there exist $n_2 \in \mathbb{N}$ and $u'_j, w'_j, b'_j \in \mathbb{F}$ for all $j \in [n_2]$ such that

$$h_2(x) = h_1(x) \oplus \bigoplus_{j=1}^{n_2} (u'_j \otimes \sigma(w'_j \otimes x \oplus b'_j)) = f^*(x)$$

for all $x \in [-1, 1]_{\mathbb{F}}$. Since h_2 can be represented by a two-layer network, choose $f = h_2$. This completes the proof.

C.2. Proof of Theorem 4

Let $0 = z_\gamma < z_{\gamma-1} < \dots < z_1 = \Omega$ be all floats between 0 and Ω , and

$$\mathcal{V} = \{2^e : e \in [\mathfrak{e}_{\min} - p, \mathfrak{e}_{\max} - p - 1]_{\mathbb{Z}}\} \cup \{(2^{\mathfrak{e}_{\max} - p - 1})^+\}.$$

In this proof, we inductively construct two-layer networks $h_1, \dots, h_\gamma$ such that

$$h_m(x) = \begin{cases} 0 & \text{if } x \in [0, z_m]_{\mathbb{F}}, \\ f^*(x) & \text{if } x \in [z_m, \Omega]_{\mathbb{F}}, \end{cases} \quad (11)$$

for all $m \in [\gamma]$. For the base case $m = 1$, we choose

$$h_1(x) = f^*(\Omega) \otimes \sigma(2^{p-\mathfrak{e}_{\max}} \otimes x \ominus (\Omega^- \times 2^{p-\mathfrak{e}_{\max}}))$$

Then, one can observe that $h_1(\Omega) = f^*(\Omega)$ and $h_1(x) = 0$ for all $x \in [0, \Omega]_{\mathbb{F}}$.

Now, we consider the general case $m > 1$. By the induction hypothesis, there exists h_{m-1} satisfying Eq. (11). Furthermore, by Lemma 24, there exist $l \in \mathbb{N}$ and $v_1, \dots, v_l \in \mathbb{F}$ such that $|f^*(z_m)| = \bigoplus_{i=1}^l v_i$. Using Lemma 23, we choose $u_i, w_i, b_i \in \mathbb{F}$ for all $i \in [l]$ such that

$$\begin{aligned} u_i \otimes \sigma(w_i \otimes x \oplus b_i) &\in \mathbb{F} & \text{if } x \in [0, z_m]_{\mathbb{F}}, \\ u_i \otimes \sigma(w_i \otimes x \oplus b_i) &= \begin{cases} v_i & \text{if } x = z_m, \\ 0 & \text{if } x \in (z_m, \Omega]_{\mathbb{F}}. \end{cases} \end{aligned} \quad (12)$$

Let $g_0 = h_{m-1}$. To construct h_m , we inductively construct f_i, g_i for $i \in [l]$ as follows: for $s \in \{-1, 1\}$ such that $s \times f^*(z_m) \geq 0$,

$$\begin{aligned} f_i(x) &= g_{i-1}(x) \oplus ((su_i) \otimes \sigma(w_i \otimes x \oplus b_i)) \\ g_i(x) &= f_i(x) \oplus \bigoplus_{k=1}^{r_i} (u'_{i,k} \otimes \sigma(w'_{i,k} \otimes x \oplus b'_{i,k})), \end{aligned}$$

where $r_i \in \mathbb{N}$ and $u'_{i,k}, w'_{i,k}, b'_{i,k} \in \mathbb{F}$ are chosen by using Lemma 25 so that

$$g_i(x) = \begin{cases} 0 & \text{if } x \in [0, z_m]_{\mathbb{F}}, \\ f_i(x) & \text{if } x \in [z_m, \Omega]_{\mathbb{F}}. \end{cases}$$

From the definitions of f_i, g_i , and by Eq. (12), one can observe that for each $i \in [l] \cup \{0\}$,

$$g_i(x) = \begin{cases} 0 & \text{if } x \in [0, z_m)_{\mathbb{F}}, \\ \bigoplus_{j=1}^i ((su_j) \otimes \sigma(w_j \otimes x \oplus b_j)) & \text{if } x = z_m, \\ f^*(x) & \text{if } x \in (z_m, \Omega]_{\mathbb{F}}. \end{cases}$$

Hence, choosing $h_m = g_l$ completes the proof.

C.3. Proof of Theorem 5

Let $g : \mathbb{F}^d \rightarrow \mathbb{F}^{2d}$ defined as $g(x) = \sigma(W \otimes x)$ for some $W \in \mathbb{F}^{2d \times d}$ so that

$$g(x_1, \dots, x_d) = (\sigma(x_1), \dots, \sigma(x_d), \sigma(-x_1), \dots, \sigma(-x_d)).$$

Then, g is injective. Choose $h^* : \mathbb{F}^{2d} \rightarrow \mathbb{F}$ such that $h^* \circ g = f^*$ on $\mathbb{F}^d$. By Theorem 7, there exists a floating-point network $h : \mathbb{F}^{2d} \rightarrow \mathbb{F}$ of width $2d + 4$ such that $h = h^*$ on $\mathbb{F}^{2d}$. Then, one can observe that $h \circ g = f^*$ on $\mathbb{F}^d$ and $h \circ g$ is a floating-point network of width $2d + 4$. This completes the proof.

C.4. Proof of Theorem 6

Let f be a floating-point network of width $\leq 2d - 1$ such that $f = f^*$ on $(-2^{e^*}, 2^{e^*})_{\mathbb{F}}$ (f^* will be decided later). Without loss of generality, we assume that f has at least two layers; otherwise, f should be coordinate-wise monotone. Let $W \otimes x \oplus b$ be the first affine transformation for some $d' \leq 2d - 1$, $W \in \mathbb{F}^{d' \times d}$, and $b = (b_1, \dots, b_{d'}) \in \mathbb{F}^{d'}$. We further assume that $|W_{ij}| \leq 2^{\epsilon_{\max} - e^* + 1}$. Otherwise, if $W_{ij} > 2^{\epsilon_{\max} - e^* + 1}$ (or $W_{ij} < -2^{\epsilon_{\max} - e^* + 1}$) for some $j \in [d]$, then by Lemma 26 and since f contains at least two layers, there exists $x \in (-2^{e^*}, 2^{e^*})_{\mathbb{F}}^d$ such that $f(x) = \text{NaN}$, or $f(x)$ is identical for all $x = (x_1, \dots, x_d) \in (-2^{e^*}, 2^{e^*})_{\mathbb{F}}^d$ with $x_j = (2^{e^*})^-$ (or $x_j = -((2^{e^*})^-)$).

For each $i \in [d]$, let $x_i, x'_i, y_i, y'_i, z_i, z'_i \in (-2^{e^*}, 2^{e^*})_{\mathbb{F}}^d$ such that

- $x_{ij} = x'_{ij} = y_{ij} = y'_{ij} = z_{ij} = z'_{ij} = 0$ for all $j < i$,
- $x_{ij} = x'_{ij} = y_{ij} = y'_{ij} = z_{ij} = z'_{ij} = (2^{e^*})^-$ for all $j > i$,
- $x_{ii} = 0, x'_{ii} = \omega, y_{ii} = 2^{\epsilon_{\min} + 4}, y'_{ii} = (2^{\epsilon_{\min} + 4})^+, z_{ii} = -2^{\epsilon_{\min} + 4}$, and $z'_{ii} = (-2^{\epsilon_{\min} + 4})^-$;

here we let $(2^{\epsilon_{\max} + 1})^- = \Omega$. Consider a target function f^* such that

$$f^*(x_i) \neq f^*(x'_i), \quad f^*(y_i) \neq f^*(y'_i), \quad \text{and} \quad f^*(z_i) \neq f^*(z'_i) \quad \text{for all } i \in [d].$$

Recall that $f = f^*$ on $(-2^{e^*}, 2^{e^*})_{\mathbb{F}}^d$. Let $W_i = (W_{i1}, \dots, W_{id})^T$ be the i -th row of W for all $i \in [d]$. In this proof, we show that if $W_k \otimes x_i \oplus b_k \neq W_k \otimes x'_i \oplus b_k$, then

- $W_k \otimes x_j = W_k \otimes x'_j, W_k \otimes y_j = W_k \otimes y'_j$, and $W_k \otimes z_j = W_k \otimes z'_j$ for all $j \neq i$, and
- $\sigma(W_k \otimes y_i \oplus b_k) = \sigma(W_k \otimes y'_i \oplus b_k)$ or $\sigma(W_k \otimes z_i \oplus b_k) = \sigma(W_k \otimes z'_i \oplus b_k)$.

Namely, even if we use d rows to separate x_i, x'_i , the pair y_i, y'_i (or z_i, z'_i) is not separated. We will also show that to separate them, additional d rows are required, which will complete the proof.

Without loss of generality, suppose that $W_1 \otimes x_i \oplus b_1 \neq W_1 \otimes x'_i \oplus b_1$; here, we also suppose $W_{1i} \in (0, 2^{\epsilon_{\max} - e^* + 1}]_{\mathbb{F}}$ by the assumption made at the beginning of this proof. Then, we must have $W_{1k} = 0$ for all $k > i$; otherwise, since $|W_{1k} \otimes (2^{e^*})^-| \geq \omega \otimes (2^{e^*})^- = (2^{e^* + \epsilon_{\min} - p})^-$ but $|W_{1i}| \otimes \omega \leq 2^{\epsilon_{\max} + \epsilon_{\min} - p - e^* + 1}$, it holds that $W_1 \otimes x_i = W_1 \otimes x'_i$ due to the round-off error and by the definition of e^* . This implies that

$$W_1 \otimes x_k = W_1 \otimes x'_k, \quad W_1 \otimes y_k = W_1 \otimes y'_k, \quad \text{and} \quad W_1 \otimes z_k = W_1 \otimes z'_k \quad \text{for all } k > i.$$

One can also observe that $W_1 \otimes x_k = W_1 \otimes x'_k$, $W_1 \otimes y_k = W_1 \otimes y'_k$, and $W_1 \otimes z_k = W_1 \otimes z'_k$ for all $k < i$ since $|W_{1i} \otimes (2^{e^*})^-| \geq (2^{e^* + \epsilon_{\min} - p})^-$ and either $|W_{1l} \otimes (2^{e^*})^-| \geq (2^{e^* + \epsilon_{\min} - p})^-$ or $W_{1l} = 0$ for all $l \in \{k+1, \dots, i-1\}$ but $|W_{1k} \otimes (2^{\epsilon_{\min} + 4})^+| \leq (2^{\epsilon_{\min} + 4 + \epsilon_{\max} - e^* + 1})^+ < 2^{e^* + \epsilon_{\min} - 2p - 2}$. Furthermore, since

$$(W_{1i} \otimes \omega) \oplus b_1 = W_1 \otimes x'_i \oplus b_1 \neq W_1 \otimes x_i \oplus b_1 = b_1,$$

we must have $|b_1| \leq 2^{\epsilon_{\min} + e_w + 3}$ where e_w is an integer such that $2^{e_w - 1} < |W_{1i}| \leq 2^{e_w}$; here, $e_w \geq 0$ since $W_{1i} \otimes \omega \neq 0$. Since $|W_{1i} \otimes (2^{\epsilon_{\min} + 4})^+| \geq 2^{\epsilon_{\min} + e_w + 3} \geq |b_1|$, this implies that $\sigma(W_1 \otimes y_i \oplus b_1) = \sigma(W_1 \otimes y'_i \oplus b_1)$ if $W_{1i} \leq 0$ and $\sigma(W_1 \otimes z_i \oplus b_1) = \sigma(W_1 \otimes z'_i \oplus b_1)$ if $W_{1i} > 0$.

So far, we proved that d rows are required to separate x_i and x'_i for all $i \in [d]$, and for each $i \in [d]$, y_i, y'_i or z_i, z'_i cannot be separated by using that d rows. Without loss of generality, we assume that y_i, y'_i are not separated for all $i \in [d]$. We now show that additional d rows are required to separate them. In addition, without loss of generality, suppose that $W_2 \otimes y_i \neq W_2 \otimes y'_i$. Then, as we previously argued, $W_{2k} = 0$ for all $k > i$ since $|W_{2i}| \otimes (2^{\epsilon_{\min} + 4})^+ \leq (2^{\epsilon_{\min} + 4 + \epsilon_{\max} - e^* + 1})^+$ but either $|W_{2k}| \otimes (2^{e^*})^- \geq (2^{e^* + \epsilon_{\min} - p})^-$ or $W_{2l} = 0$. Likewise, we have $W_2 \otimes y_k = W_2 \otimes y'_k$ for all $k < i$ since $|W_{2i} \otimes (2^{e^*})^-| \geq (2^{e^* + \epsilon_{\min} - p})^-$ and either $|W_{2l} \otimes (2^{e^*})^-| \geq (2^{e^* + \epsilon_{\min} - p})^-$ or $W_{2k} = 0$ for all $l \in \{k+1, \dots, i-1\}$ but $|W_{2k} \otimes (2^{\epsilon_{\min} + 4})^+| \leq (2^{\epsilon_{\min} + 4 + \epsilon_{\max} - e^* + 1})^+ < 2^{e^* + \epsilon_{\min} - 2p - 2}$. Hence, y_k, y'_k cannot be separated by using W_2, b_2 for all $k \neq i$. This completes the proof.

C.5. Proof of Theorem 9

Let $d = a \times (p+1) + b$ for some non-negative integers a and $b \in [0, p]_{\mathbb{Z}}$. Let $g : \mathbb{F}^d \rightarrow \mathbb{F}^{\lceil d/(p+1) \rceil_{\mathbb{Z}}}$ defined as $g(x) = \sigma(W \otimes x)$ for some $W \in \mathbb{F}^{\lceil d/(p+1) \rceil_{\mathbb{Z}} \times d}$ so that

$$g(x_1, \dots, x_d) = \left(\bigoplus_{i=1}^{p+1} (2^{i-1} \otimes x_i), \dots, \bigoplus_{i=1}^{p+1} (2^{i-1} \otimes x_{(a-1)(p+1)+i}), \bigoplus_{i=1}^b (2^{i-1} \otimes x_{a(p+1)+i}) \right)$$

where the last coordinate exists if and only if $b \geq 1$. Then, g is injective. Choose $h^* : \mathbb{F}^{\lceil d/(p+1) \rceil_{\mathbb{Z}}} \rightarrow \mathbb{F}$ such that $h^* \circ g = f^*$ on $\mathbb{F}^d$. By Theorem 7, there exists a floating-point network $h : \mathbb{F}^{\lceil d/(p+1) \rceil_{\mathbb{Z}}} \rightarrow \mathbb{F}$ of width $\lceil d/(p+1) \rceil_{\mathbb{Z}} + 4$ such that $h = h^*$ on $\mathbb{F}^{\lceil d/(p+1) \rceil_{\mathbb{Z}}}$. Then, one can observe that $h \circ g = f^*$ on $\mathbb{F}^d$ and $h \circ g$ is a floating-point network of width $\lceil d/(p+1) \rceil_{\mathbb{Z}} + 4$. This completes the proof.

C.6. Proof of Lemma 10

Without loss of generality, assume $s = 1$. The case $s = -1$ follows similarly.

First, consider the case $|z| \geq 1$. In this case, we choose $w = -2^{-2}$ and $b = z \times 2^{-2}$. Then, for $x \geq z$,

$$\sigma(w \otimes x \oplus b) = 0,$$

and for $x < z$,

$$\Omega \times 2^{-1} \geq \Omega \times 2^{-2} \oplus |\Omega \times 2^{-2}| \geq \sigma(w \otimes x \oplus b) \geq (z - z^-) \times 2^{-2} \geq 2^{-3-p} \geq \omega.$$

By choosing $u = -1$, we obtain the desired conclusion.

Next, consider the case $|z| < 1$. In this case, we choose $w = -1$ and $b = z$. Then, for $x \geq z$,

$$\sigma(w \otimes x \oplus b) = 0,$$

and for $x < z$,

$$\Omega \geq \Omega \oplus |1| \geq \sigma(w \otimes x \oplus b) \geq (z - z^-) > 0.$$

By choosing $u = -1$, we obtain the desired conclusion. This completes the proof.

C.7. Proof of Lemma 11

Lemma 27. For any $k \in [\epsilon_{\min} - p, \epsilon_{\max} - p - 1]_{\mathbb{Z}}$, there exists $0 < \alpha \in \mathbb{F}$ such that $2^k \oplus \alpha \oplus (-\alpha) = 2^{k+1}$.

Proof. Choose $\alpha = (2^{k+p+1})^+$. then, $2^k \oplus \alpha = (2^{k+p+1})^{++}$. Hence, $2^k \oplus \alpha \oplus (-\alpha) = 2^{k+1}$. □

By Lemma 27, there exist $m \in \mathbb{N}$ and $0 < \beta_1, \dots, \beta_m \in \mathbb{F}$ such that $\omega \oplus \beta_1 \oplus (-\beta_1) \oplus \dots \oplus \beta_m \oplus (-\beta_m) = 1$. Choose $n = 2m$, $\alpha_{2i} = -\beta_i$, and $\alpha_{2i-1} = \beta_i$ for all $i \in [m]$. Then, one can observe that all conditions in Lemma 11 are satisfied. This completes the proof.